Integrated NSM and Threat Hunting Laboratory

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Suricata + Zeek + RITA Implementation with Attack Simulation and Correlati

Executive Summary

This laboratory exercise successfully designed, deployed, and validated a complete Network Security Monitoring (NSM) and Threat Hunting pipeline using **Suricata**, **Zeek**, and **RITA** (with MongoDB). The environment consisted of a dedicated sensor (Ubuntu) and an attacker machine (Kali) to simulate realistic adversarial activity.

Three controlled attacks were executed:

- Beaconing-style repeated connections
- Aggressive port scanning
- DNS anomaly / tunneling simulation

Key Results:

- Suricata generated strong real-time alerts, including **2107+ NMap SYN Scan** detections and **55 DNS Tunneling Attempt** alerts.
- Zeek successfully captured detailed connection, DNS, and HTTP metadata for forensic analysis.
- RITA effectively identified **exploded DNS activity** on domains such as evil-c2.com and malicious-domain.com, along with multiple beacon candidates.

The lab demonstrated a full detection-to-hunting workflow: real-time alerting (Suricata) → rich network logging (Zeek) → behavioral threat hunting (RITA). This validates the effectiveness of combining signature-based detection with behavioral analytics for modern SOC operations.

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VM's Network Setup & Guideline

VM	Role	Tools Installed	Purpose
Ubuntu 192.168.100.142	Sensor / Monitoring Machine	Suricata + Zeek + MongoDB + Rita	Capture, detect, hunt, and investigate
Kali 192.168.100.140	Attacker / Traffic Generator	Basic tools (nmap, curl, python scripts, etc.)	Generate normal + malicious traffic cleanly

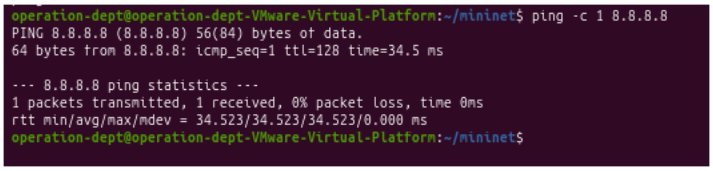
PART 1: SURICATA — IDS/IPS ENGINE

Network Security Monitoring & Threat Hunting Pipeline

STEP 1: PRE-INSTALLATION SETUP

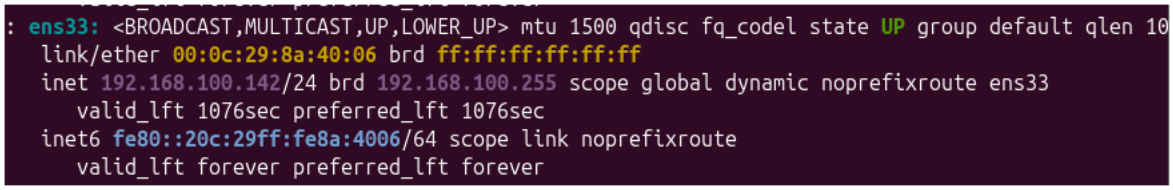
1.1 Verify System Requirements

```
ping -c 1 8.8.8.8
```



Check available network interfaces

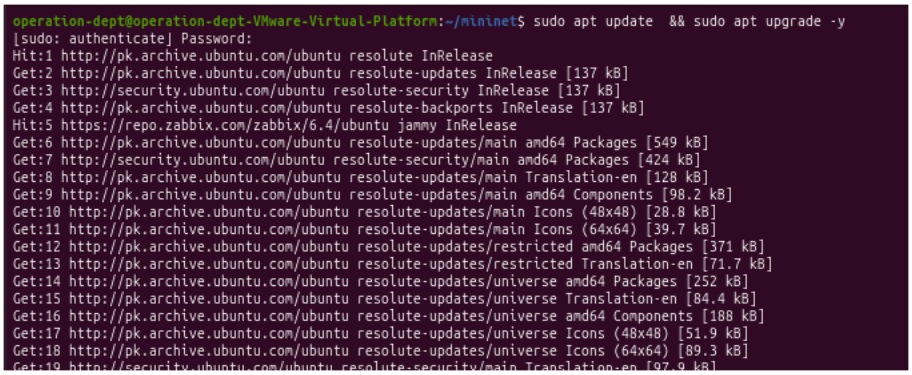
```
ip a
```



1.2 Update System Packages

Update package manager

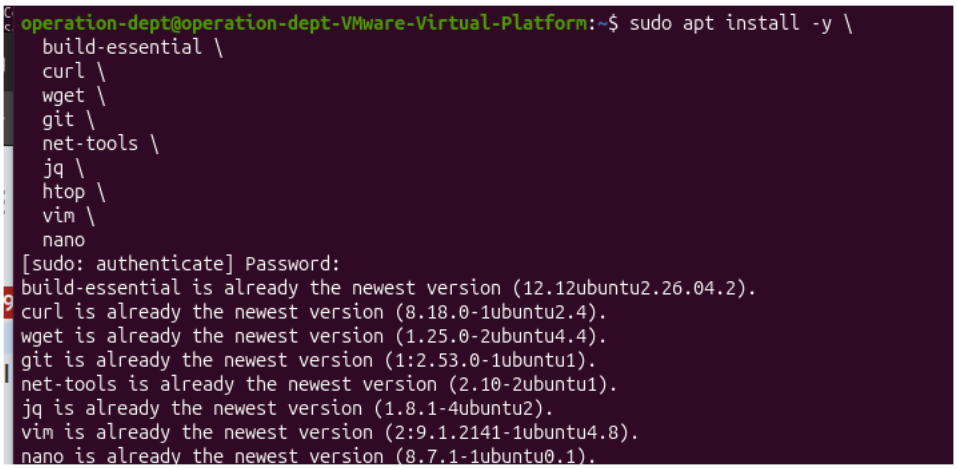
```
sudo apt update && sudo apt upgrade -y
```



Install essential build tools (needed for some components)

```
sudo apt install -y \
  build-essential \
```

```
curl \
wget \
git \
net-tools \
jq \
htop \
vim \
nano
```

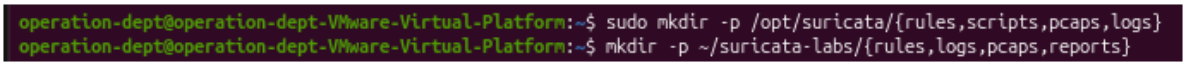


1.3 Create Working Directories

Create directories for Suricata logs and configs & also a user working directory

```
sudo mkdir -p /opt/suricata/{rules,scripts,pcaps,logs}
```

```
mkdir -p ~/suricata-labs/{rules,logs,pcaps,reports}
```



Set proper permissions

```
sudo chown $USER:$USER ~/suricata-labs
```

```
sudo chmod 755 ~/suricata-labs
```



STEP 2: SURICATA INSTALLATION & VERIFICATION

2.1 Install Suricata from Ubuntu Repository

Install Suricata package

```
sudo apt install -y suricata
```



Verify installation

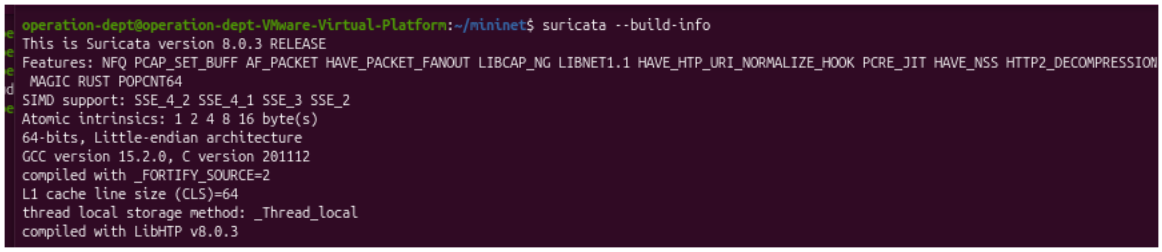
```
suricata –version
```



2.2 Get Full Build Information

Display comprehensive build information

```
suricata --build-info
```



```
operation-dept@operation-dept-VMware-Virtual-Platform:~/mininet$ suricata --build-info
This is Suricata version 8.0.3 RELEASE
Features: NFQ PCAP_SET_BUFF AF_PACKET HAVE_PACKET_FANOUT LIBCAP_NG LIBNET1.1 HAVE_HTP_URI_NORMALIZE_HOOK PCRE_JIT HAVE_NSS HTTP2_DECOMPRESSION HA
 MAGIC RUST POPCNT64
SIMD support: SSE_4_2 SSE_4_1 SSE_3 SSE_2
Atomic intrinsics: 1 2 4 8 16 byte(s)
64-bits, Little-endian architecture
GCC version 15.2.0, C version 201112
compiled with _FORTIFY_SOURCE=2
L1 cache line size (CLS)=64
thread local storage method: _Thread_local
compiled with LibHTP v8.0.3

Suricata Configuration:
  AF_PACKET support:                       yes
  AF_XDP support:                          yes
  DPDK support:                            yes
  eBPF support:                            yes
  XDP support:                             yes
  PF_RING support:                         no
  NFQueue support:                         yes
  NFLOG support:                           yes
  IPFW support:                            no
  Netmap support:                          no
  DAG enabled:                             no
  Napatech enabled:                        no
  WinDivert enabled:                       no
  Npcap support:

  Unix socket enabled:                     yes
  Detection enabled:                       yes

  Libmagic support:                        yes
  libjansson support:                      yes
  hiredis support:                         yes
  hiredis async with libevent:             yes
  PCRE jit:                                yes
  GeoIP2 support:                          yes
  JA3 support:                             yes
  JA4 support:                             yes
  Hyperscan support:                       yes
  Hwloc support:                           yes
  Libnet support:                          yes
  liblz4 support:                          yes
  Landlock support:                        yes
  Systemd support:                         yes

  Rust strict mode:                        no
  Rust compiler path:                      /usr/bin/rustc
```

2.3 Verify Key Configuration Files

Verify main config file exists

```
ls -la /etc/suricata/suricata.yaml
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/mininet$ ls -la /etc/suricata/suricata.yaml
-rw-r--r-- 1 root root 93083 Jan 13  2026 /etc/suricata/suricata.yaml
```

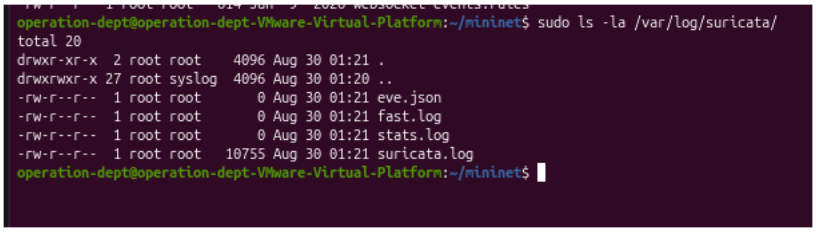
Verify rules directory exists

```
ls -la /etc/suricata/rules/
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/mininet$ ls -la /etc/suricata/rules/
total 176
drwxr-xr-x 2 root root  4096 Aug 30 01:20 .
drwxr-xr-x 3 root root  4096 Aug 30 01:20 ..
-rw-r--r-- 1 root root  1858 Jan  9  2026 app-layer-events.rules
-rw-r--r-- 1 root root 21287 Jan  9  2026 decoder-events.rules
-rw-r--r-- 1 root root   468 Jan  9  2026 dhcp-events.rules
-rw-r--r-- 1 root root  1786 Jan  9  2026 dnp3-events.rules
-rw-r--r-- 1 root root  2202 Jan  9  2026 dns-events.rules
-rw-r--r-- 1 root root   524 Jan  9  2026 enip-events.rules
-rw-r--r-- 1 root root  4005 Jan  9  2026 files.rules
-rw-r--r-- 1 root root   446 Jan  9  2026 ftp-events.rules
-rw-r--r-- 1 root root 15088 Jan  9  2026 http-events.rules
-rw-r--r-- 1 root root  3907 Jan  9  2026 http2-events.rules
-rw-r--r-- 1 root root  2834 Jan  9  2026 ipsec-events.rules
-rw-r--r-- 1 root root   585 Jan  9  2026 kerberos-events.rules
-rw-r--r-- 1 root root  2576 Jan  9  2026 mdns-events.rules
-rw-r--r-- 1 root root  1856 Jan  9  2026 modbus-events.rules
-rw-r--r-- 1 root root  2187 Jan  9  2026 mqtt-events.rules
-rw-r--r-- 1 root root   729 Jan  9  2026 nfs-events.rules
-rw-r--r-- 1 root root   558 Jan  9  2026 ntp-events.rules
-rw-r--r-- 1 root root   658 Jan  9  2026 pgsql-events.rules
-rw-r--r-- 1 root root   882 Jan  9  2026 pop3-events.rules
-rw-r--r-- 1 root root   721 Jan  9  2026 quic-events.rules
-rw-r--r-- 1 root root   926 Jan  9  2026 rfb-events.rules
-rw-r--r-- 1 root root  4607 Jan  9  2026 smb-events.rules
-rw-r--r-- 1 root root  5433 Jan  9  2026 smtp-events.rules
-rw-r--r-- 1 root root   719 Jan  9  2026 ssh-events.rules
-rw-r--r-- 1 root root 17569 Jan  9  2026 stream-events.rules
-rw-r--r-- 1 root root  7081 Jan  9  2026 tls-events.rules
-rw-r--r-- 1 root root   614 Jan  9  2026 websocket-events.rules
```

Verify log directory exists

```
sudo ls -la /var/log/suricata/
```



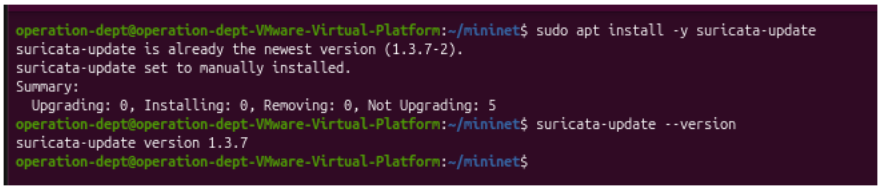
STEP 3: RULE MANAGEMENT

3.1 Install suricata-update Tool

Install rule update utility & verify installation

```
sudo apt install -y suricata-update
```

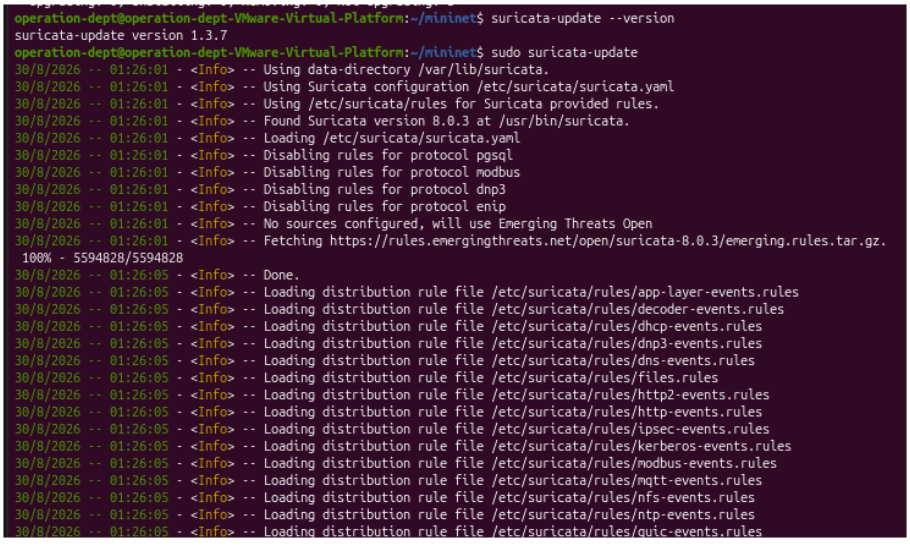
```
suricata-update –version
```



3.2 Download Emerging Threats Rules (First Time)

Download and update rules from Emerging Threats Open

```
sudo suricata-update
```



3.3 Verify Rules Downloaded Successfully

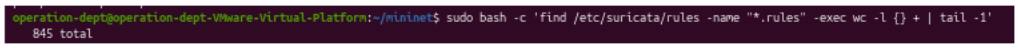
Count rules in each file

```
sudo bash -c 'for file in /etc/suricata/rules/*.rules; do echo "$file: $(grep -c "^alert\|^drop\|^pass" "$file") rules"; done'
```



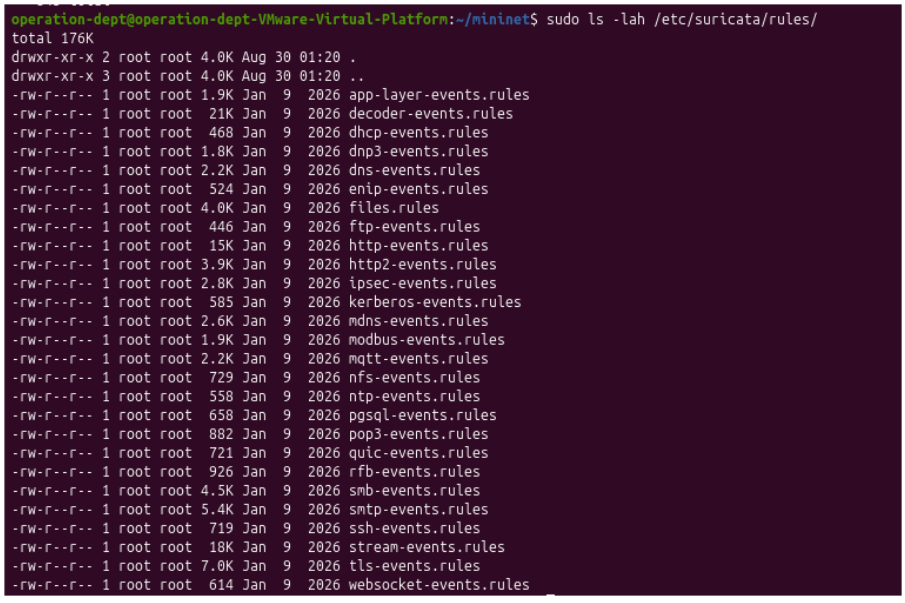
Show total rules available

```
sudo bash -c 'find /etc/suricata/rules -name "*.rules" -exec wc -l {} + | tail -1'
```



List all rule files

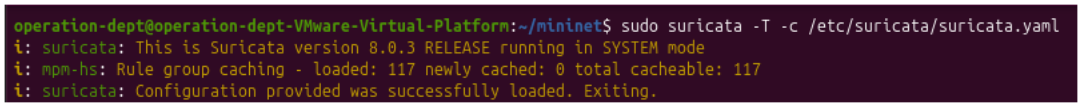
```
sudo ls -lah /etc/suricata/rules/
```



3.4 Test Configuration Before Live Running

Test that all rules load correctly (CRITICAL STEP)

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```



STEP 4: CONFIGURATION TUNING

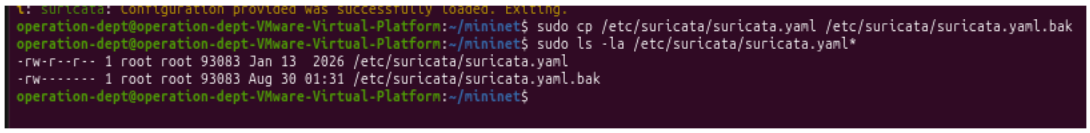
4.1 Backup Original Configuration

Always backup before editing critical files

```
sudo cp /etc/suricata/suricata.yaml /etc/suricata/suricata.yaml.bak
```

Verify backup

```
sudo ls -la /etc/suricata/suricata.yaml*
```



4.2 Edit Main Configuration File

Open configuration in nano editor

```
sudo nano /etc/suricata/suricata.yaml
```

```
form:~/mininet$ sudo nano /etc/suricata/suricata.yaml
```

Find these sections and verify/update:

Section 1: Network Interfaces

```
# Linux high speed capture support
af-packet:
  - interface: ens33
    # Number of receive threads. "auto" uses the number of cores
    #threads: auto
    # Default clusterid. AF_PACKET will load balance packets based on flow.
    cluster-id: 99
    # Default AF_PACKET cluster type. AF_PACKET can load balance per flow or per hash.
    # This is only supported for Linux kernel > 3.1
    # possible value are:
    #  * cluster_flow: all packets of a given flow are sent to the same socket
    #  * cluster_cpu: all packets treated in kernel by a CPU are sent to the same socket
    #  * cluster_qm: all packets linked by network card to a RSS queue are sent to the same
    #  socket. Requires at least Linux 3.14.
    #  * cluster_ebpf: eBPF file load balancing. See doc/userguide/capture-hardware/ebpf-xdp.rst
    #  more info.
    # Recommended modes are cluster_flow on most boxes and cluster_cpu or cluster_qm on system
    # with capture card using RSS (requires cpu affinity tuning and system IRQ tuning)
    # cluster_rollover has been deprecated; if used, it'll be replaced with cluster_flow.
    cluster-type: cluster_flow
```

Section 2: Rule Files

Find rule-files: section - Should look like this:

```
##
## Configure Suricata to load Suricata-Update managed rules.
##

default-rule-path: /etc/suricata/rules

rule-files:
  - local.rules

##
## Auxiliary configuration files.
##

classification-file: /etc/suricata/classification.config
reference-config-file: /etc/suricata/reference.config
# threshold-file: /etc/suricata/threshold.config

##
## Suricata as a Firewall options (experimental)
##
firewall:
  # toggle to enable firewall mode
  #enabled: no

  # Firewall rule files are in their own path and are not managed
```

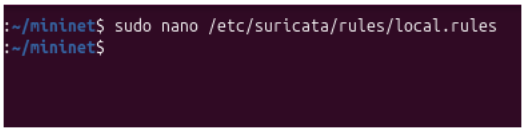
IMPORTANT: Ensure - local.rules is listed (we will add custom rules there).

STEP 5: CREATE LOCAL RULES FILE

5.1 Create local.rules File

Create local rules file for custom rules

```
sudo nano /etc/suricata/rules/local.rules
```



5.2 Verify local.rules is Empty but Valid

Check file was created

```
sudo ls -la /etc/suricata/rules/local.rules
```



Verify it's readable

```
sudo cat /etc/suricata/rules/local.rules
```



STEP 6: CUSTOM RULE WRITING (STANDARD + ADVANCED)

6.1 Add IMPORTANT Custom Rules

Edit local.rules

```
sudo nano /etc/suricata/rules/local.rules
```

Paste ALL these rules into the file:

```
# ==== SECTION 1: MALWARE & C2 DETECTION ====
alert tcp any any -> any any (msg:"Suspicious PowerShell Download"; flow:established; content:"powershell"; nocase; content:"-Command"; nocase; sid:9000001; rev:2;)
alert http any any -> any any (msg:"Known Malware User-Agent Detected"; flow:established,to_server; http.user_agent; pcre:"/(WinHTTP|Cvk8rowser|Nmap|Masscan)/i"; sid:9000002; rev:2;)
alert dns any any -> any any (msg:"Known Malicious Domain Query"; dns.query; content:"malware.local"; nocase; sid:9000003; rev:2;)
alert http any any -> any any (msg:"Emotet C2 Beaconing Pattern"; flow:established,to_server; http.method; content:"POST"; http.user_agent; content:"Mozilla/5.0 (Windows NT"; nocase; sid:9000004; rev:2;)

# ==== SECTION 2: RECONNAISSANCE & SCANNING ====
alert tcp any any -> any any (msg:"Possible NMap SYN Scan Detected"; flags:S; flow:stateless; sid:9000005; rev:2;)
alert udp any any -> any any (msg:"UDP Port Scan Detected"; threshold:type both,track by_src,count 25,seconds 10; sid:9000006; rev:2;)
alert http any any -> any any (msg:"SQL Injection Attack Detected"; flow:established,to_server; http.method; content:"GET"; http.uri; content:"union"; nocase; content:"select"; nocase; sid:9000007; rev:2;)
alert http any any -> any any (msg:"Directory Traversal Attack"; flow:established,to_server; http.uri; content:"../"; sid:9000008; rev:2;)
alert http any any -> any any (msg:"NMap HTTP Probe Detected"; flow:established,to_server; http.user_agent; content:"Nmap"; nocase; sid:9000009; rev:2;)

# ==== SECTION 3: DATA EXFILTRATION ====
alert http any any -> any any (msg:"Large HTTP Response - Possible Data Exfiltration"; flow:established,to_client; http.content_len; byte_test:4,>,5000000,0; sid:9000010; rev:2;)
alert dns any any -> any any (msg:"DNS Tunneling Attempt Detected"; dns.query; pcre:"/^.{50,}$/"; sid:9000011; rev:2;)
alert tcp any any -> any any (msg:"SSH Traffic Detected"; flow:established; content:"SSH-"; depth:4; sid:9000012; rev:2;)
```

```
# ==== SECTION 4: PROTOCOL ANOMALIES ====
alert tcp any any -> any [!80,!8080,!8888] (msg:"HTTP Traffic on Non-Standard Port"; content:"GET"; nocase; offset:0; depth:3; sid:9000013; rev:2;)
alert tcp any any -> any [!443,!8443] (msg:"TLS Traffic on Non-Standard Port"; content:"|16 03|"; offset:0; depth:2; sid:9000014; rev:2;)
alert dns any any -> any any (msg:"Excessive NXDOMAIN Responses - Possible DGA Malware"; dns.rcode:3; threshold:type both,track by_src,count 50,seconds 30; sid:9000015; rev:2;)
alert dns any any -> any any (msg:"DNS Query for Tor Hidden Service"; dns.query; content:".onion"; nocase; sid:9000016; rev:2;)

# ==== SECTION 5: CUSTOM WEB FILTERING ====
drop http any any -> any any (msg:"YouTube Access Blocked"; flow:established,to_server; http.host; content:"youtube"; sid:9000017; rev:2; metadata:policy security;)
drop http any any -> any any (msg:"Facebook Access Blocked"; flow:established,to_server; http.host; content:"facebook"; sid:9000018; rev:2; metadata:policy security;)
drop http any any -> any any (msg:"Reddit Access Blocked"; flow:established,to_server; http.host; content:"reddit"; sid:9000019; rev:2; metadata:policy security;)
alert http any any -> any any (msg:"Twitter Access Detected"; flow:established,to_server; http.host; content:"twitter.com"; sid:9000020; rev:2;)

# ==== SECTION 6: ADVANCED BEHAVIORAL DETECTION ====
alert tcp !192.168.0.0/16 any -> 192.168.0.0/16 [22,3389] (msg:"Suspicious SSH/RDP Connection from External Network"; flags:S; sid:9000021; rev:2;)
alert http any any -> any any (msg:"HTTP Basic Auth Attempt Detected"; flow:established,to_server; http.request_header; content:"Authorization:"; nocase; content:"Basic"; nocase; threshold:type both,track by_src,count 10,seconds 60; sid:900
0022; rev:2;)
alert udp any any -> any 88 (msg:"Possible Kerberos Brute Force"; threshold:type both,track by_src,count 20,seconds 30; sid:9000023; rev:2;)
alert icmp any any -> any any (msg:"Possible ICMP Sweep Detected"; threshold:type both,track by_src,count 20,seconds 60; sid:9000024; rev:3;)
```

6.2 Verify Rules Saved

```
sudo wc -l /etc/suricata/rules/local.rules
```

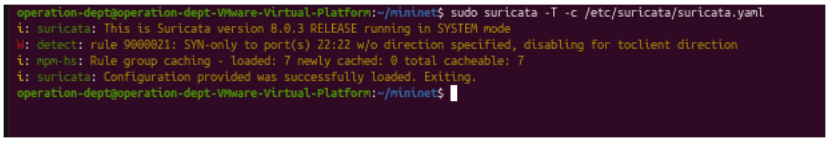
```
operation-dept@operation-dept-VMware-Virtual-Platform:~/mininet$ sudo wc -l /etc/suricata/rules/local.rules
73 /etc/suricata/rules/local.rules
operation-dept@operation-dept-VMware-Virtual-Platform:~/mininet$
```

STEP 7: TEST ALL RULES LOAD

7.1 Validate All Rules (Including Custom)

Test entire configuration with custom rules

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```



STEP 8: CONFIGURE SURICATA SERVICE

8.1 Enable Suricata to Start on Boot

Enable Suricata service

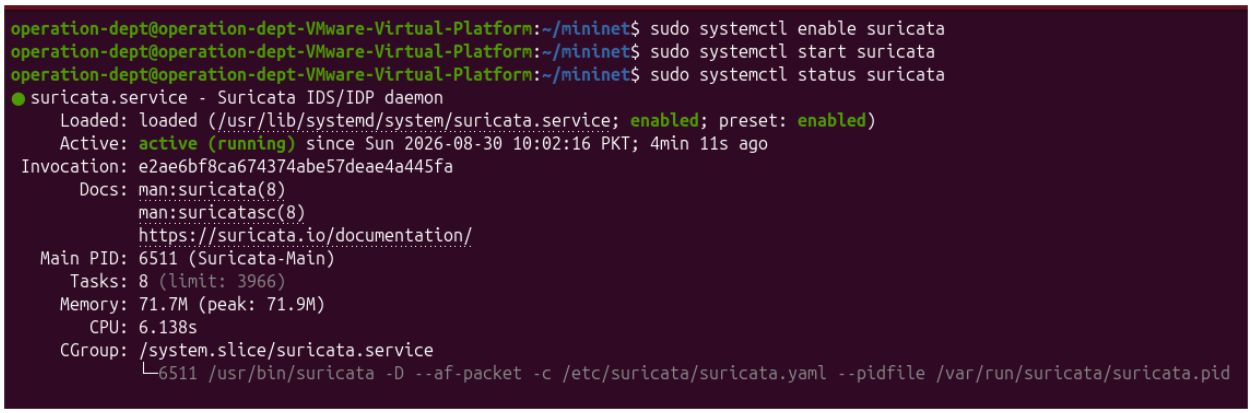
```
sudo systemctl enable suricata
```

Start Suricata service

```
sudo systemctl start suricata
```

Verify service is running

```
sudo systemctl status suricata
```



STEP 9: Test IDS Mode (Alert Generation)

9.1 Monitor Alerts in Real-Time

Open Terminal 1 on Sensor (Ubuntu 192.168.100.142):

```
sudo tail -f /var/log/suricata/fast.log
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/mininet$ sudo tail -f /var/log/suricata/fast.log
08/30/2026-10:11:26.923467  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:34630 -> 172.64.144.52:443
08/30/2026-10:11:26.980647  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 172.64.144.52:443 -> 192.168.100.142:34630
08/30/2026-10:11:26.981704  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 172.64.144.52:443 -> 192.168.100.142:34614
08/30/2026-10:11:26.981707  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 172.64.144.52:443 -> 192.168.100.142:34598
08/30/2026-10:12:04.261226  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:43874 -> 34.54.185.247:443
08/30/2026-10:12:04.505040  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.54.185.247:443 -> 192.168.100.142:43874
08/30/2026-10:12:04.509057  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.54.185.247:443 -> 192.168.100.142:43874
08/30/2026-10:12:04.628581  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.54.185.247:443 -> 192.168.100.142:43874
08/30/2026-10:12:07.366468  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:34110 -> 151.101.141.91:80
08/30/2026-10:12:07.456697  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:34870 -> 140.248.129.91:80
```

9.2 Generate Test Traffic

Open Terminal 2 on Sensor or Attacker (Kali 192.168.100.140):

Generate HTTP traffic to trigger rules:

Test 1: Normal Google Traffic

```
curl http://www.google.com -v
```

```
┌──(crack㉿kali)-[~]
└─$ curl http://www.google.com -v
* Host www.google.com:80 was resolved.
* IPv6: 2001:4860:482b:7700::, 2001:4860:4827:7700::, 2001:4860:482a:7700::, 2001:4860:482d:7700::, 2001:
4860:4829:7700::, 2001:4860:482c:7700::, 2001:4860:4828:7700::, 2001:4860:4826:7700::
* IPv4: 142.251.151.119, 142.251.152.119, 142.251.153.119, 142.251.157.119, 142.251.154.119, 142.251.156.
119, 142.251.150.119, 142.251.155.119
*   Trying [2001:4860:482b:7700::]:80 ...
* Immediate connect fail for 2001:4860:482b:7700::: Network is unreachable
* connect to 2001:4860:482b:7700:: port 80 from :: port 0 failed: Network is unreachable
*   Trying 142.251.151.119:80 ...
* Established connection to www.google.com (142.251.151.119 port 80) from 192.168.100.140 port 42996
* using HTTP/1.x
> GET / HTTP/1.1
> Host: www.google.com
> User-Agent: curl/8.21.0
```

Test 2: Simulate YouTube access

```
curl -H "Host: youtube.com" http://www.google.com/ -v
```

```
┌──(crack㉿kali)-[~]
└─$ curl -H "Host.:youtube.com" http://www.google.com/ -v
* Host www.google.com:80 was resolved.
* IPv6: 2001:4860:4826:7700::, 2001:4860:4828:7700::, 2001:4860:482c:7700::, 2001:4860:4829:7700::, 2001:
4860:482d:7700::, 2001:4860:482a:7700::, 2001:4860:4827:7700::, 2001:4860:482b:7700::
* IPv4: 142.251.155.119, 142.251.150.119, 142.251.156.119, 142.251.154.119, 142.251.157.119, 142.251.153.
119, 142.251.152.119, 142.251.151.119
*   Trying [2001:4860:4826:7700::]:80 ...
* Immediate connect fail for 2001:4860:4826:7700::: Network is unreachable
* connect to 2001:4860:4826:7700:: port 80 from :: port 0 failed: Network is unreachable
*   Trying 142.251.155.119:80 ...
* Established connection to www.google.com (142.251.155.119 port 80) from 192.168.100.140 port 55508
* using HTTP/1.x
> GET / HTTP/1.1
> Host: www.google.com
> User-Agent: curl/8.21.0
> Accept: */*
```

Test 3: Simulate Facebook access

```
curl -H "Host: facebook.com" http://www.google.com/ -v
```



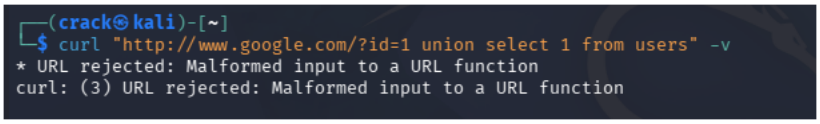
Test 4: Simulate NMap tool

```
curl -A "Nmap Scripting Engine" http://www.google.com/ -v
```



Test 5: Simulate SQL injection

```
curl "http://www.google.com/?id=1 union select 1 from users" -v
```



9.3 Watch Alerts Appear

In Terminal 1, you should see alerts like:

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/mininet$ sudo tail -f /var/log/suricata/fast.log
[sudo: authenticate] Password:
08/30/2026-10:28:49.877873  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:50394 -> 91.189.91.113:4460
08/30/2026-10:28:50.164369  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 91.189.91.113:4460 -> 192.168.100.142:50394
08/30/2026-10:28:50.164416  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:50394 -> 91.189.91.113:4460
08/30/2026-10:28:50.165999  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 91.189.91.113:4460 -> 192.168.100.142:50394
08/30/2026-10:28:50.205739  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:41768 -> 91.189.91.111:4460
08/30/2026-10:28:50.423684  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:41768 -> 91.189.91.111:4460
08/30/2026-10:28:50.636350  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 91.189.91.111:4460 -> 192.168.100.142:41768
08/30/2026-10:28:50.636403  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:41768 -> 91.189.91.111:4460
08/30/2026-10:28:50.637193  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 91.189.91.111:4460 -> 192.168.100.142:41768
08/30/2026-10:28:55.819972  [**] [1:9000006:2] UDP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {UDP} 192.168.100.142:39157 -> 91.189.91.113:123
08/30/2026-10:30:27.646147  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:59464 -> 142.251.153.119:80
08/30/2026-10:30:27.888055  [**] [1:9000002:2] Known Malware User-Agent Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:59464 -> 142.251.153.119:80
08/30/2026-10:30:27.888055  [**] [1:9000009:2] NMap HTTP Probe Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:59464 -> 142.251.153.119:80
08/30/2026-10:30:32.922946  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:59266 -> 142.251.155.119:80
08/30/2026-10:30:58.980683  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:40194 -> 142.251.157.119:80
08/30/2026-10:31:06.478988  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:40202 -> 142.251.157.119:80

08/30/2026-10:31:44.138805  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:50812 -> 142.251.153.119:80
08/30/2026-10:31:44.374572  [**] [1:9000002:2] Known Malware User-Agent Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:50812 -> 142.251.153.119:80
08/30/2026-10:31:44.374572  [**] [1:9000009:2] NMap HTTP Probe Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:50812 -> 142.251.153.119:80
08/30/2026-10:31:52.184770  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:46434 -> 142.251.157.119:80
08/30/2026-10:33:48.296324  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:33004 -> 185.125.190.101:80
08/30/2026-10:34:20.643052  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:46778 -> 140.248.129.91:80
08/30/2026-10:34:21.501877  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:37360 -> 34.149.226.178:443
08/30/2026-10:34:21.523913  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:37362 -> 34.149.226.178:443
08/30/2026-10:34:21.701970  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:34450 -> 151.101.141.91:443
08/30/2026-10:34:21.710648  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:37374 -> 34.149.226.178:443
08/30/2026-10:34:21.826185  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37362
08/30/2026-10:34:21.821867  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 151.101.141.91:443 -> 192.168.100.142:34450
08/30/2026-10:34:21.836227  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37362
08/30/2026-10:34:21.831386  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37360
08/30/2026-10:34:21.838822  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37360
08/30/2026-10:34:21.951081  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37362
08/30/2026-10:34:21.955595  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37360
```

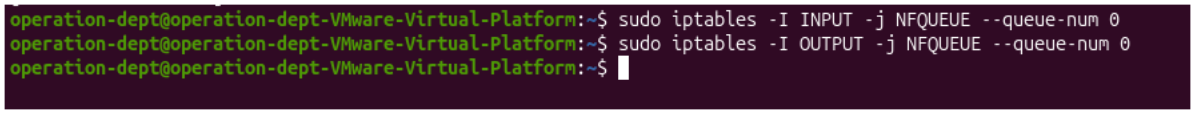
```
08/30/2026-10:31:44.138805  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:50812 -> 142.251.153.119:80
08/30/2026-10:31:44.374572  [**] [1:9000002:2] Known Malware User-Agent Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:50812 -> 142.251.153.119:80
08/30/2026-10:31:44.374572  [**] [1:9000009:2] NMap HTTP Probe Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:50812 -> 142.251.153.119:80
08/30/2026-10:31:52.184770  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.140:46434 -> 142.251.157.119:80
08/30/2026-10:33:48.296324  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:33004 -> 185.125.190.101:80
08/30/2026-10:34:20.643052  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:46778 -> 140.248.129.91:80
08/30/2026-10:34:21.501877  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:37360 -> 34.149.226.178:443
08/30/2026-10:34:21.523913  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:37362 -> 34.149.226.178:443
08/30/2026-10:34:21.701970  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:34450 -> 151.101.141.91:443
08/30/2026-10:34:21.710648  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:37374 -> 34.149.226.178:443
08/30/2026-10:34:21.826185  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37362
08/30/2026-10:34:21.821867  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 151.101.141.91:443 -> 192.168.100.142:34450
08/30/2026-10:34:21.836227  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37362
08/30/2026-10:34:21.831386  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37360
08/30/2026-10:34:21.838822  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37360
08/30/2026-10:34:21.951081  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37362
08/30/2026-10:34:21.955595  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37360
08/30/2026-10:34:22.018176  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37374
08/30/2026-10:34:22.020810  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37374
08/30/2026-10:34:22.138593  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.149.226.178:443 -> 192.168.100.142:37374
08/30/2026-10:34:22.248497  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:34452 -> 151.101.141.91:443
08/30/2026-10:34:22.320829  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 151.101.141.91:443 -> 192.168.100.142:34452
08/30/2026-10:34:22.324702  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 151.101.141.91:443 -> 192.168.100.142:34452
08/30/2026-10:34:22.358691  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 151.101.141.91:443 -> 192.168.100.142:34452
08/30/2026-10:34:22.388830  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 151.101.141.91:443 -> 192.168.100.142:34452
08/30/2026-10:34:22.413929  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:34462 -> 151.101.141.91:443
08/30/2026-10:34:22.519257  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 151.101.141.91:443 -> 192.168.100.142:34462
08/30/2026-10:34:24.236624  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:35392 -> 34.107.243.93:443
08/30/2026-10:34:24.609799  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.107.243.93:443 -> 192.168.100.142:35392
08/30/2026-10:34:24.735996  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:35404 -> 34.107.243.93:443
08/30/2026-10:34:24.990703  [**] [1:9000006:2] UDP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {UDP} 192.168.100.142:60340 -> 192.168.100.2:53
08/30/2026-10:34:24.993560  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:36840 -> 34.54.185.247:443
08/30/2026-10:34:24.994162  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:46788 -> 140.248.129.91:80
08/30/2026-10:34:25.027358  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.107.243.93:443 -> 192.168.100.142:35404
08/30/2026-10:34:25.078844  [**] [1:9000006:2] UDP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {UDP} 192.168.100.2:53 -> 192.168.100.142:53142
08/30/2026-10:34:25.348857  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 34.54.185.247:443 -> 192.168.100.142:36840
08/30/2026-10:35:08.807433  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:57910 -> 104.18.32.47:443
08/30/2026-10:35:08.858404  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 104.18.32.47:443 -> 192.168.100.142:57910
08/30/2026-10:35:10.372651  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:59904 -> 150.171.109.163:443
08/30/2026-10:35:10.486424  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 150.171.109.163:443 -> 192.168.100.142:59904
08/30/2026-10:35:16.631017  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:57388 -> 13.35.171.34:443
08/30/2026-10:35:16.746300  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 13.35.171.34:443 -> 192.168.100.142:57388
08/30/2026-10:35:20.106339  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:52594 -> 172.64.148.235:443
08/30/2026-10:35:20.157159  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 172.64.148.235:443 -> 192.168.100.142:52594
08/30/2026-10:35:20.186061  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:52602 -> 172.64.148.235:443
08/30/2026-10:35:20.207061  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:34246 -> 140.248.129.91:80
08/30/2026-10:35:20.220795  [**] [1:9000014:2] TLS Traffic on Non-Standard Port [**] [Classification: (null)] [Priority: 3] {TCP} 172.64.148.235:443 -> 192.168.100.142:52602
08/30/2026-10:35:20.457213  [**] [1:9000005:2] Possible NMap SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.100.142:34260 -> 140.248.129.91:80
```

STEP 10: Test IPS Mode (Packet Blocking)

10.1 Setup NFQUEUE (Packet Interception)

```
sudo iptables -I INPUT -j NFQUEUE --queue-num 0
```

```
sudo iptables -I OUTPUT -j NFQUEUE --queue-num 0
```



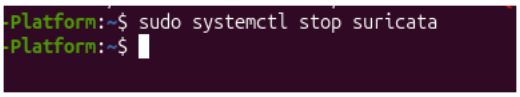
10.2 Verify iptables Rules Added

```
sudo iptables -L -n -v | grep NFQUEUE
```



10.3 Stop Suricata (IDS Mode)

```
sudo systemctl stop suricata
```



9.4 Start Suricata in IPS Mode

```
sudo suricata -c /etc/suricata/suricata.yaml -q 0 -v
```



9.5 Test Blocking YouTube (from another terminal)

Try to access YouTube (should timeout/fail - packet dropped)

```
curl -H "Host: youtube.com" http://www.google.com/ --max-time 5
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ curl -H "Host: youtube.com" http://www.google.com/ --max-time 5
curl: (28) Operation timed out after 5003 milliseconds with 0 bytes received
operation-dept@operation-dept-VMware-Virtual-Platform:~$
```

Expected Result: Connection timeout (packet was dropped)

9.6 Clean Up IPS Mode

Press Ctrl+C in Suricata terminal

```
^CNotice: suricata: Signal Received.  Stopping engine.
Info: suricata: time elapsed 229.740s
Notice: nfq: (RX-NFQ#0) Treated: Pkts 2567, Bytes 839826, Errors 0
Notice: nfq: (RX-NFQ#0) Verdict: Accepted 2473, Dropped 94, Replaced 0
Info: counters: Alerts: 104
operation-dept@operation-dept-VMware-Virtual-Platform:~$
```

e

Remove iptables rules

```
sudo iptables -D INPUT -j NFQUEUE --queue-num 0
```

```
sudo iptables -D OUTPUT -j NFQUEUE --queue-num 0
```

Verify rules removed

```
sudo iptables -L -n -v | grep NFQUEUE
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ sudo iptables -D INPUT -j NFQUEUE --queue-num 0
sudo iptables -D OUTPUT -j NFQUEUE --queue-num 0
operation-dept@operation-dept-VMware-Virtual-Platform:~$
operation-dept@operation-dept-VMware-Virtual-Platform:~$ sudo iptables -L -n -v | grep NFQUEUE
    5   772 NFQUEUE    all  --  *      *       0.0.0.0/0            0.0.0.0/0            NFQUEUE num 0
   62 52677 NFQUEUE    all  --  *      *       0.0.0.0/0            0.0.0.0/0            NFQUEUE num 0
operation-dept@operation-dept-VMware-Virtual-Platform:~$
```

Restart Suricata in IDS mode

```
sudo systemctl start suricata
```

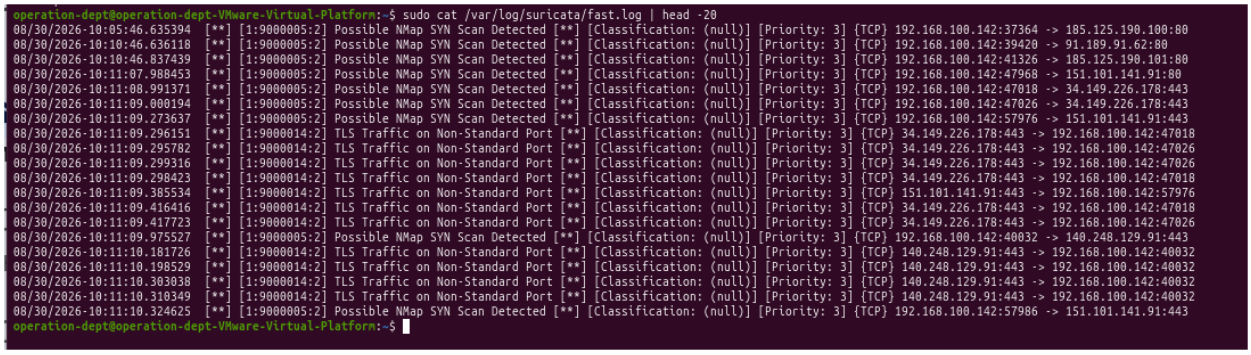
```
sudo systemctl status suricata
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ sudo systemctl start suricata
operation-dept@operation-dept-VMware-Virtual-Platform:~$ sudo systemctl status suricata
● suricata.service - Suricata IDS/IDP daemon
     Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-08-30 10:49:46 PKT; 4s ago
 Invocation: a3af6cac501149c7a1fc0ebe73e44626
       Docs: man:suricata(8)
             man:suricatasc(8)
             https://suricata.io/documentation/
   Main PID: 7743 (Suricata-Main)
      Tasks: 8 (limit: 1588)
     Memory: 42.7M (peak: 43.2M)
        CPU: 453ms
     CGroup: /system.slice/suricata.service
             └─7743 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /var/run/suricata/suricata.pid
```

STEP 10: Analyze Alerts

10.1 View Fast Log

```
sudo cat /var/log/suricata/fast.log | head -20
```



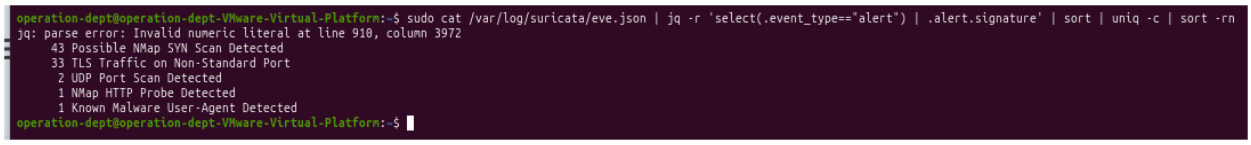
10.2 View JSON Alerts (if jq installed)

```
sudo cat /var/log/suricata/eve.json | jq 'select(.event_type=="alert")' |
head -20
```



10.3 Count Alerts by Rule

```
sudo cat /var/log/suricata/eve.json | jq -r 'select(.event_type=="alert") |
.alert.signature' | sort | uniq -c | sort -rn
```



PART 2: ZEEK — NETWORK SECURITY MONITOR

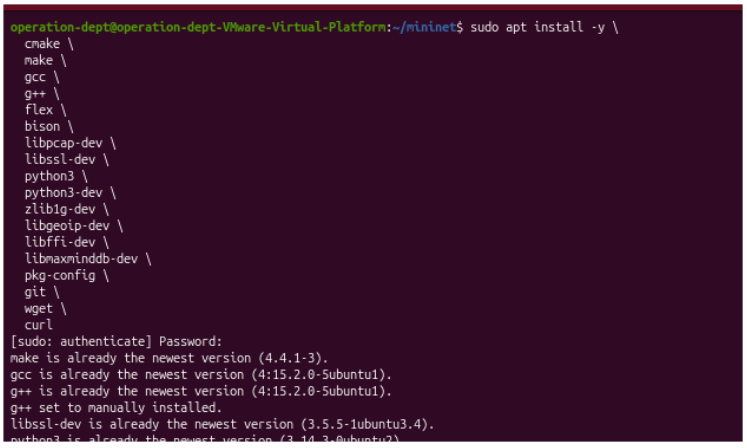
Tool	Purpose	Output	Use Case
Suricata	Threat Detection (Rules-based)	Alerts when malicious patterns match	Real-time threat blocking
Zeek	Network Analysis (Behavioral)	Logs ALL traffic details (HTTP, DNS, SSL, Flows)	Threat hunting & forensics

ZEEK INSTALLATION FROM SOURCE

2.1 Install Build Dependencies

Install all required build tools

```
sudo apt install -y \
  cmake \
  make \
  gcc \
  g++ \
  flex \
  bison \
  libpcap-dev \
  libssl-dev \
  python3 \
  python3-dev \
  zlib1g-dev \
  libgeoip-dev \
  libffi-dev \
  libmaxminddb-dev \
  pkg-config \
  git \
  wget \
  curl
```



Required libraries

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$ sudo apt update
sudo apt install -y libzmq3-dev
[sudo: authenticate] Password:
Hit:1 http://pk.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://pk.archive.ubuntu.com/ubuntu resolute-backports InRelease
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$ dpkg -l | grep libzmq
ii  cppzmq-dev:amd64                        4.10.0-1build2                  amd64        C++ bindings for libzmq (headers)
ii  libzmq3-dev:amd64                       4.3.5-1build3                   amd64        lightweight messaging kernel (development files)
ii  libzmq5:amd64                           4.3.5-1build3                   amd64        lightweight messaging kernel (shared library)
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$
```

Installing swig

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$ sudo apt install -y \
  bison cmake cppzmq-dev gcc g++ flex libfl-dev \
  libpcap-dev libssl-dev make python3 python3-dev \
  swig zlib1g-dev
[sudo: authenticate] Password:
bison is already the newest version (2:3.8.2+dfsg-1build4).
cmake is already the newest version (4.2.3-2ubuntu2).
cppzmq-dev is already the newest version (4.10.0-1build2).
cppzmq-dev set to manually installed.
gcc is already the newest version (4:15.2.0-5ubuntu1).
g++ is already the newest version (4:15.2.0-5ubuntu1).
flex is already the newest version (2.6.4-8.2build2).
```

Verify installation:

```
gcc --version

cmake --version

flex --version

bison --version
```

```
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

cmake version 4.2.3

CMake suite maintained and supported by Kitware (kitware.com/cmake).
flex 2.6.4
bison (GNU Bison) 3.8.2
Written by Robert Corbett and Richard Stallman.

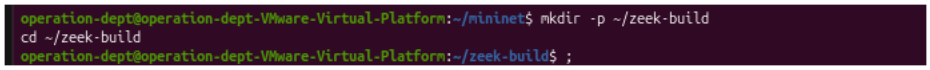
Copyright (C) 2021 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
operation-dept@operation-dept-VMware-Virtual-Platform:~/mininet$
```

2.2 Download Zeek Source Code

Create working directory

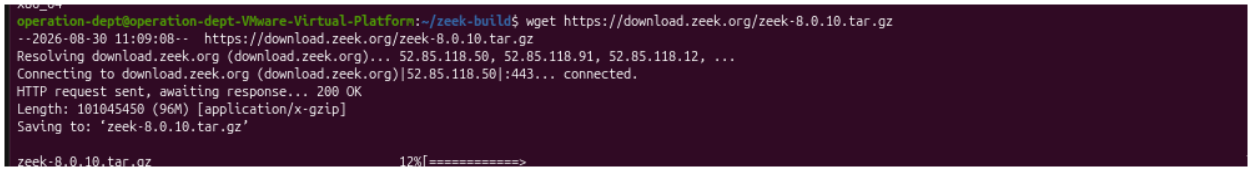
```
mkdir -p ~/zeek-build

cd ~/zeek-build
```



Download latest Zeek

```
wget https://download.zeek.org/zeek-8.0.10.tar.gz
```



Verify download



Extract



2.3 Configure Zeek Build

Create build directory

```
mkdir build

cd build
```



Configure with production options

```
cmake -DCMAKE_BUILD_TYPE=Release \
  -DCMAKE_INSTALL_PREFIX=/opt/zeek \
  -DENABLE_MOBILE_SUPPORT=yes \
  -DENABLE_PYTHON=yes \
  ..
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$ cmake -DCMAKE_BUILD_TYPE=Release \
  -DCMAKE_INSTALL_PREFIX=/opt/zeek \
  -DENABLE_MOBILE_SUPPORT=yes \
  -DENABLE_PYTHON=yes \
  ..
-- The C compiler identification is GNU 15.2.0
-- The CXX compiler identification is GNU 15.2.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Performing Test test_arch_x64
-- Performing Test test_arch_x64 - Success
-- Performing Test test_arch_aarch64
-- Performing Test test_arch_aarch64 - Failed
-- Performing Test test_arch_arm
-- Performing Test test_arch_arm - Failed
-- Performing Test test_arch_power
-- Performing Test test_arch_power - Failed
-- Determined target architecture (for hashing): x86_64
-- Performing Test cxx_std_works
-- Performing Test cxx_std_works - Success
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Success
-- Found Threads: TRUE
-- Found sed: /usr/bin/sed
-- Found Python: /usr/bin/python3 (found suitable version "3.14.4", minimum required is "3.9.0") found components: Interpreter
-- Found FLEX: /usr/bin/flex (found version "2.6.4")
-- Found BISON: /usr/bin/bison (found suitable version "3.8.2", minimum required is "2.5")
```

```
Fuzz Targets:             OFF
Fuzz Engine:

External Tools/Linters:
  - Include What You Use: OFF
  - Clang-Tidy:           OFF

-----------------------------------------------------------
-- Configuring done (62.5s)
-- Generating done (1.8s)
CMake Warning:
  Manually-specified variables were not used by the project:

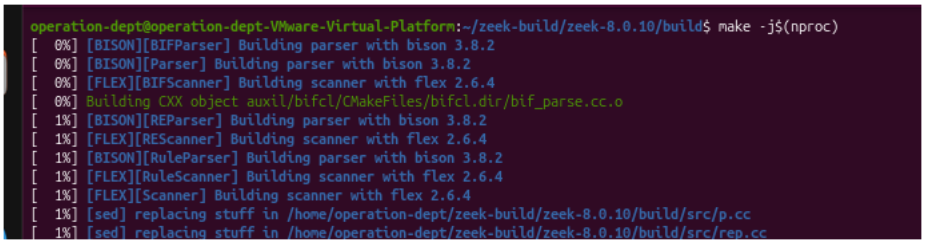
    ENABLE_MOBILE_SUPPORT
    ENABLE_PYTHON


-- Build files have been written to: /home/operation-dept/zeek-build/zeek-8.0.10/build
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$
```

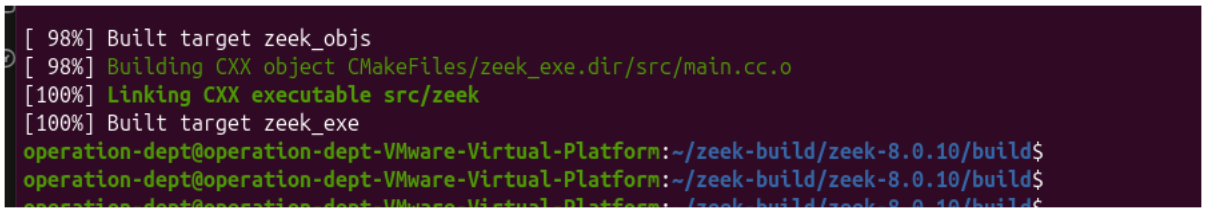
2.4 Compile Zeek (Takes 10-15 minutes)

Start compilation

```
make -j$(nproc)
```



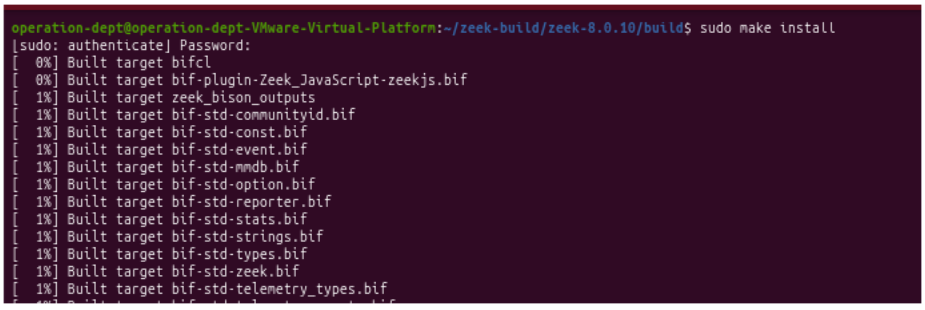
`make -j4` -> perform test validation



2.5 Install Zeek

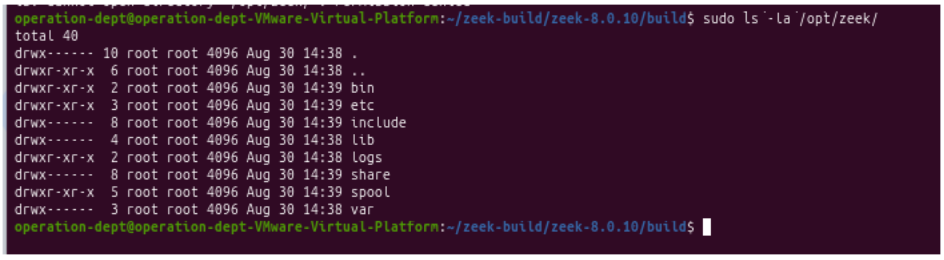
Install to /opt/zeek (production location)

```
sudo make install
```



Verify installation

```
ls -la /opt/zeek/
```



Create profile script

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$ sudo tee /etc/profile.d/zeek.sh <<EOF
export PATH=/opt/zeek/bin:\$PATH
EOF
export PATH=/opt/zeek/bin:$PATH
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$
```

s

Grant permission

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$ sudo chown root:root /etc/profile.d/zeek.sh
sudo chmod 644 /etc/profile.d/zeek.sh
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$
```

Adding it in our personal bash & load It

```
ild$ echo 'export PATH=/opt/zeek/bin:$PATH' >> ~/.bashrc
ild$
```

Verify Zeek is accessible

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$ export PATH=/opt/zeek/bin:$PATH
hash -r
zeek --version
zeek version 8.0.10
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$
```

2.6 Set Up Zeek Node Configuration (Standalone)

- Simpler and more reliable for labs
- No need for PF_RING
- Easier to troubleshoot
- Used in real-world deployments before moving to cluster mode

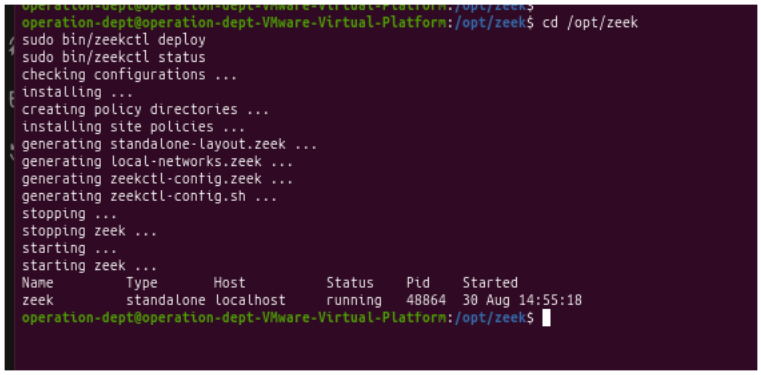
Create node configuration

```
sudo tee /opt/zeek/etc/node.cfg <<EOF
[zeek]
type=standalone
host=localhost
interface=ens33
EOF
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek$ sudo tee /opt/zeek/etc/node.cfg <<EOF
[zeek]
type=standalone
host=localhost
interface=ens33
EOF
[zeek]
type=standalone
host=localhost
interface=ens33
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek$
```

2.7 Deploy Zeek

```
cd /opt/zeek
sudo bin/zeekctl deploy
sudo bin/zeekctl status
```



ZEEK LOG CONFIGURATION

3.1 Enable JSON Logging

```
sudo tee -a /opt/zeek/share/zeek/site/local.zeek <<EOF
@load policy/tuning/json-logs
EOF
```

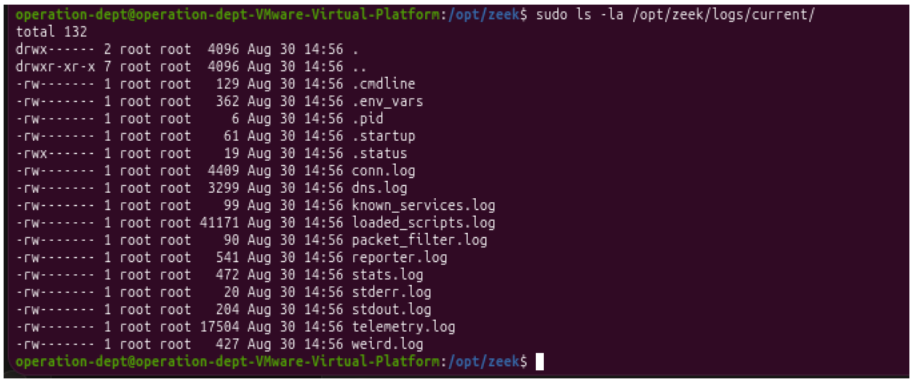
Redeploy:

```
cd /opt/zeek
sudo bin/zeekctl deploy
```



3.2 Check Logs

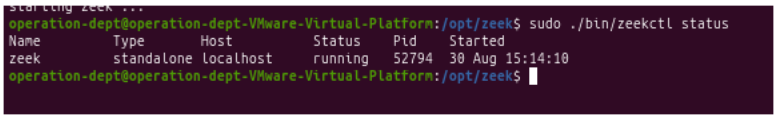
```
sudo ls -la /opt/zeek/logs/current/
```



3.3 Check if Zeek Processes are Running

```
cd /opt/zeek
```

```
sudo ./bin/zeekctl status
```



Create Logs Directory

Create the logs directory

```
sudo mkdir -p /opt/zeek/logs/current
```

Set permissions

```
sudo chown -R root:root /opt/zeek/logs
```

```
sudo chmod 755 /opt/zeek/logs
```

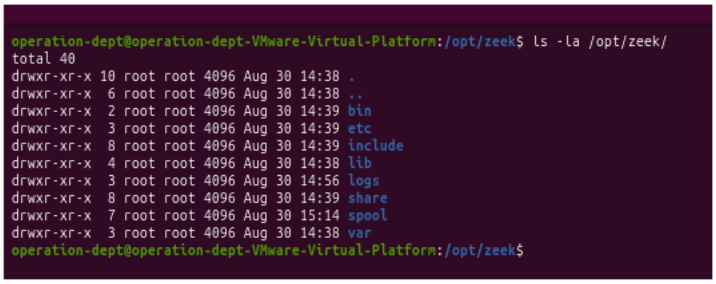
```
sudo chmod 755 /opt/zeek/logs/current
```



Verify Zeek Base Directory Structure

Check what exists

```
ls -la /opt/zeek/
```

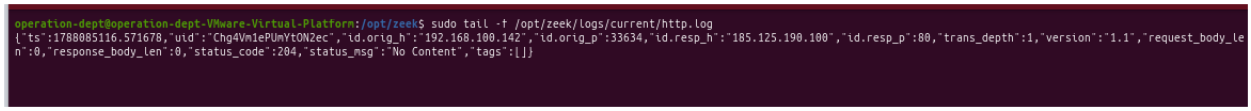


Expected: You should see bin/, etc/, share/, logs/

3.4 Generate Test Traffic to Create Logs

Terminal 1 - Monitor HTTP Logs:

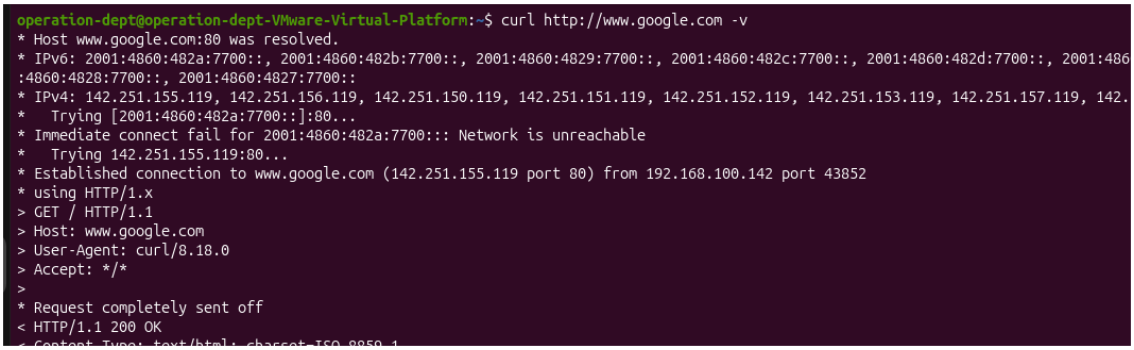
```
sudo tail -f /opt/zeek/logs/current/http.log
```



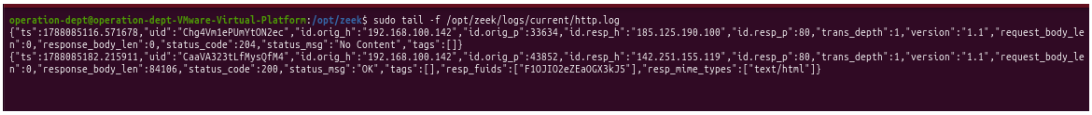
Terminal 2 - Generate Traffic:

Access Google

```
curl http://www.google.com -v
```



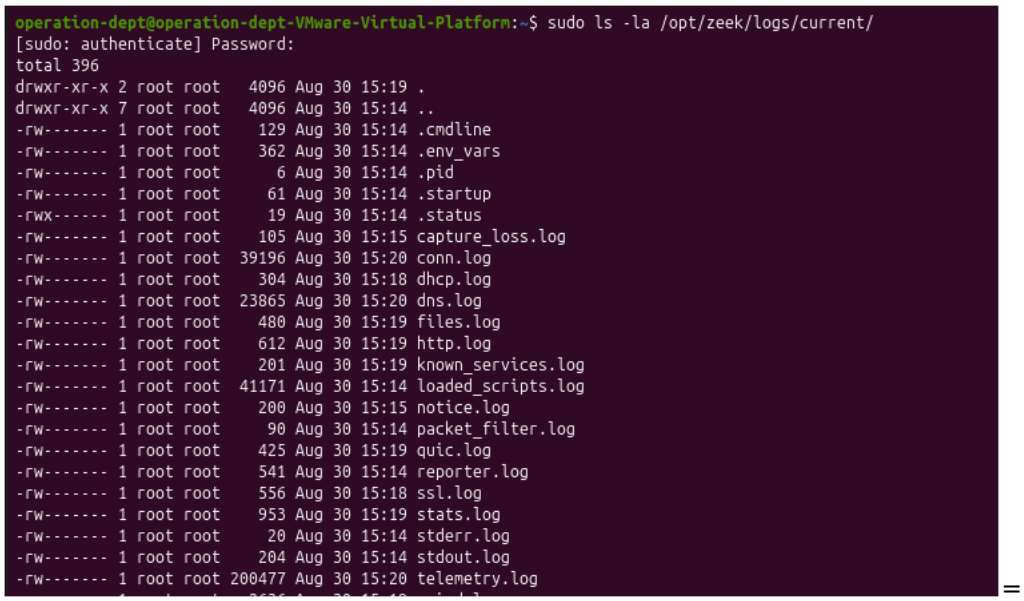
Expected: Logs should appear in Terminal 1



3.5 Verify All Log Files are Being Created

List all current logs

```
sudo ls -la /opt/zeek/logs/current/
```



=

Count logs

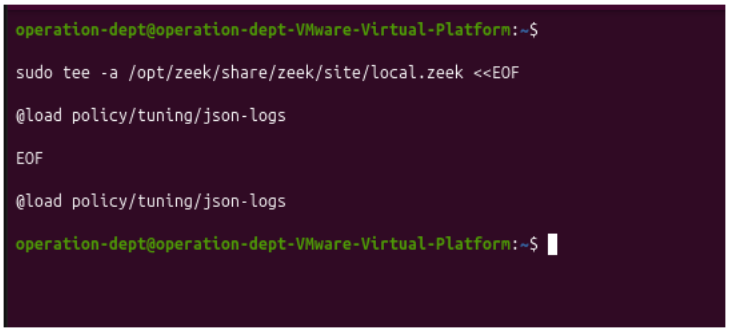
```
ls -1 /opt/zeek/logs/current/ | wc -l
```



ENABLE JSON LOGGING

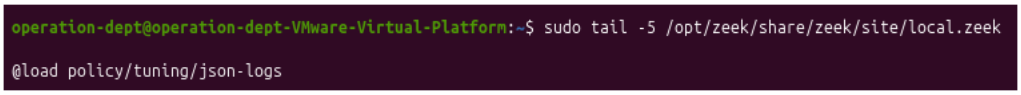
4.1 Add JSON Logging to Local Configuration

```
sudo tee -a /opt/zeek/share/zeek/site/local.zeek <<EOF
@load policy/tuning/json-logs
EOF
```



Verify it was added:

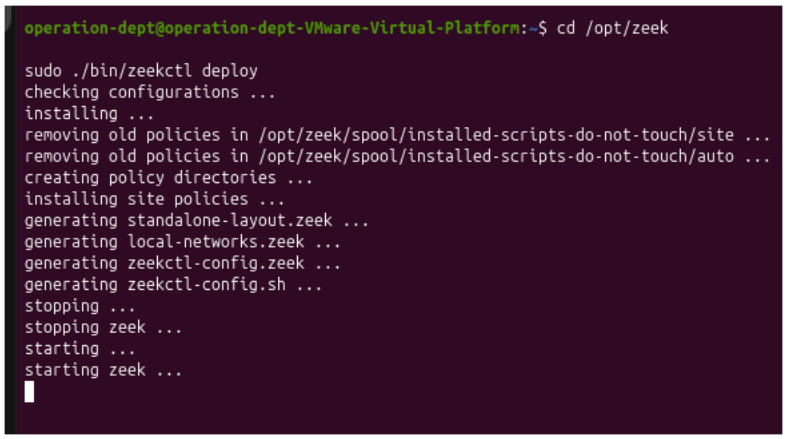
```
sudo tail -5 /opt/zeek/share/zeek/site/local.zeek
```



4.2 Redeploy Zeek with JSON Configuration

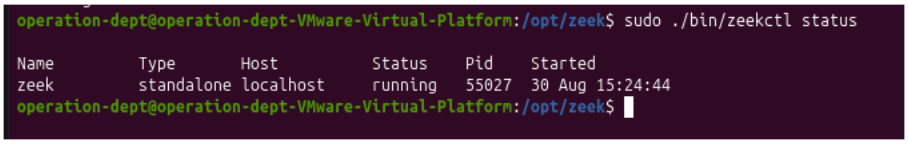
Redeploy to apply JSON logging

```
cd /opt/zeek
sudo ./bin/zeekctl deploy
```



Verify deployment successful

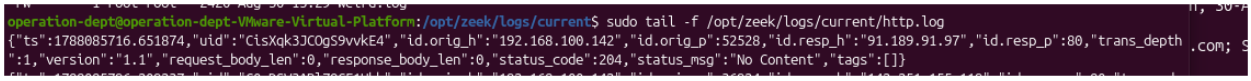
```
sudo ./bin/zeekctl status
```



4.3 Generate Test Traffic Again to Verify JSON

Terminal 1:

```
sudo tail -f /opt/zeek/logs/current/http.log
```

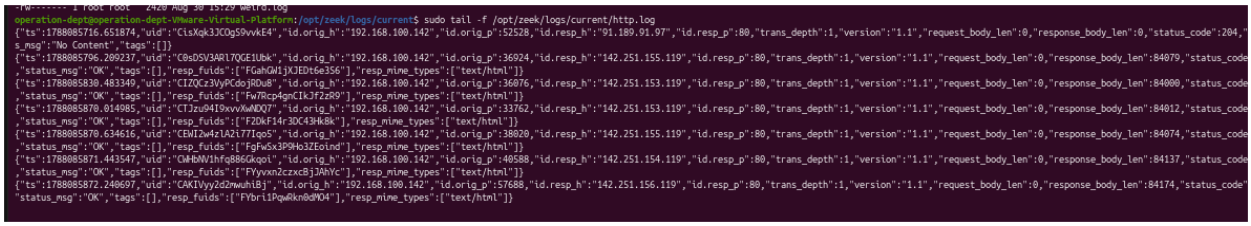


Terminal 2:

```
curl http://www.google.com -v
```



Expected: Logs appear in JSON format (with fields separated by tabs or JSON structure)



Part-3 End-to-End Attack Simulation & Investigation

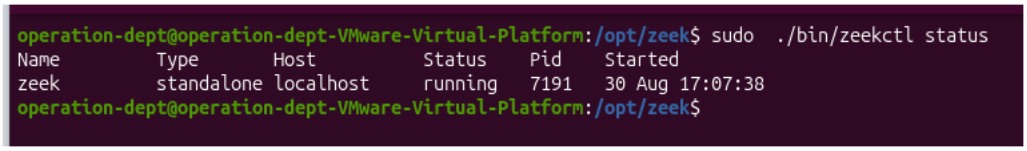
On Ubuntu Sensor (192.168.100.142)

1.1 Check both tools are running

```
sudo systemctl status suricata --no-pager
```




```
cd /opt/zeek && sudo ./bin/zeekctl status
```

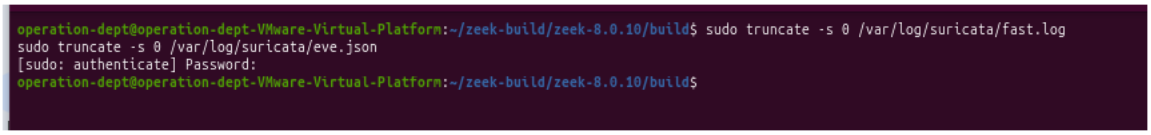


1.2 Clear old logs (recommended for clean evidence)

Suricata

```
sudo truncate -s 0 /var/log/suricata/fast.log
```

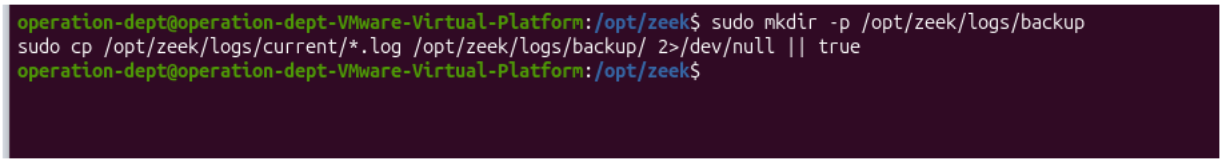
```
sudo truncate -s 0 /var/log/suricata/eve.json
```



Zeek (backup first)

```
sudo mkdir -p /opt/zeek/logs/backup
```

```
sudo cp /opt/zeek/logs/current/*.log /opt/zeek/logs/backup/ 2>/dev/null ||
true
```

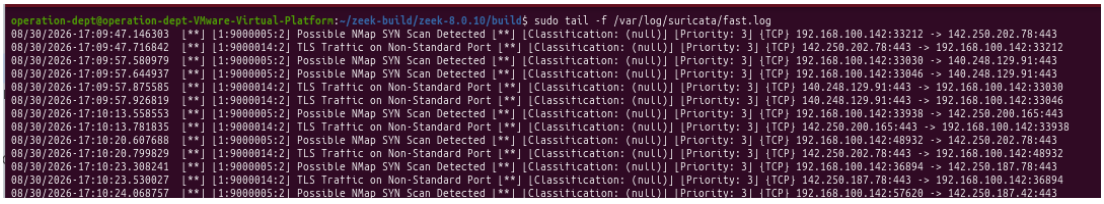


1.3 Open monitoring terminals

Open **two** terminals on Ubuntu:

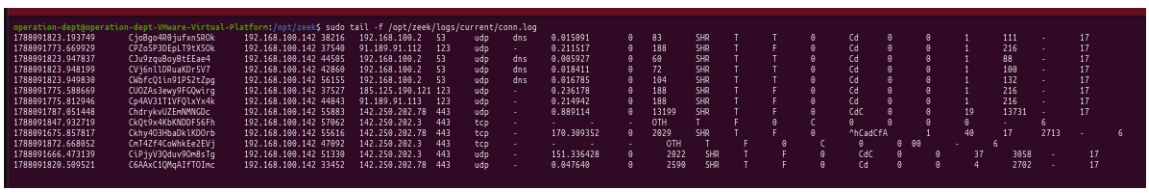
Terminal A (Suricata):

```
sudo tail -f /var/log/suricata/fast.log
```



Terminal B (Zeek):

```
sudo tail -f /opt/zeek/logs/current/conn.log
```



Leave both running.

Attack Simulations (From Kali)

Go to your **Kali machine (192.168.100.140)** and run the attacks one by one.

Attack 1: Port Scan

```
nmap -sS -T4 192.168.100.142
```

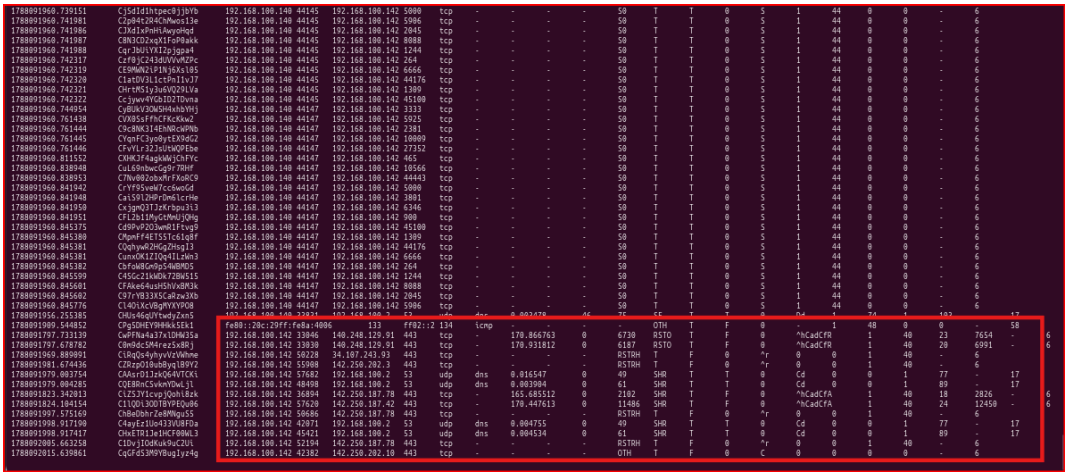
```
┌──(crack㉿kali)-[~]
└─$ nmap -sS -T4 192.168.100.142
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-30 17:12 +0500
Nmap scan report for 192.168.100.142 (192.168.100.142)
Host is up (0.0018s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: 00:0C:29:8A:40:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 4.92 seconds
```

- Suricata terminal (Terminal A)



- Zeek terminal (Terminal B)



Attack 1: Port Scan

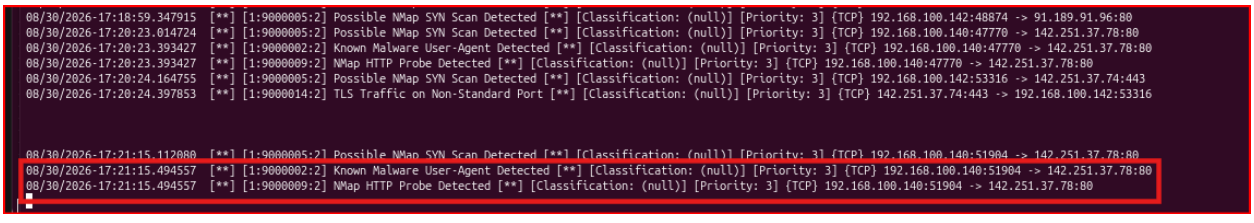
- **Suricata Alert:** Multiple “Possible Nmap SYN Scan” alerts from 192.168.100.140
- **Zeek Context:** Large number of short-lived TCP connections with S0 state (SYN sent, no response)
- **Conclusion:** Clear reconnaissance activity. Source IP 192.168.100.140 performed a SYN scan against the sensor.

Attack 2: Suspicious User-Agent

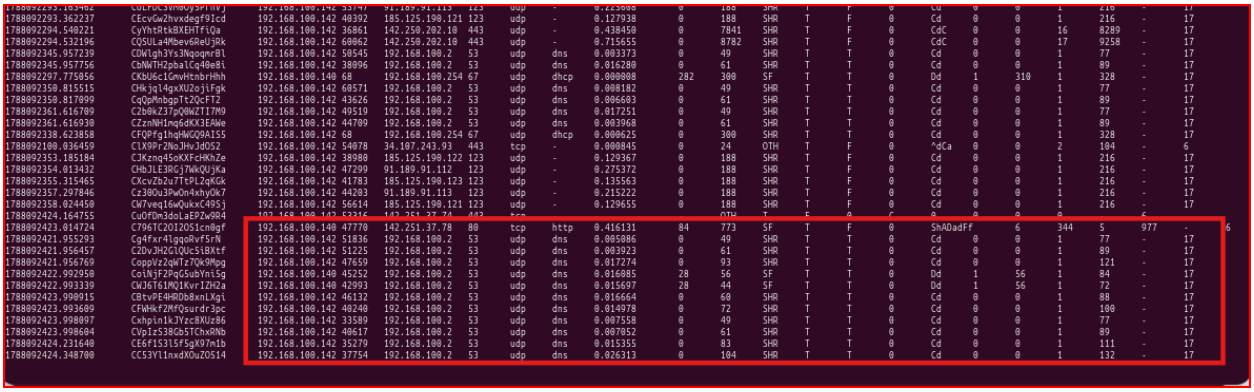
curl -A "Nmap Scripting Engine" http://google.com/ -v



- Suricata terminal (Terminal A)



- Zeek terminal (Terminal B)



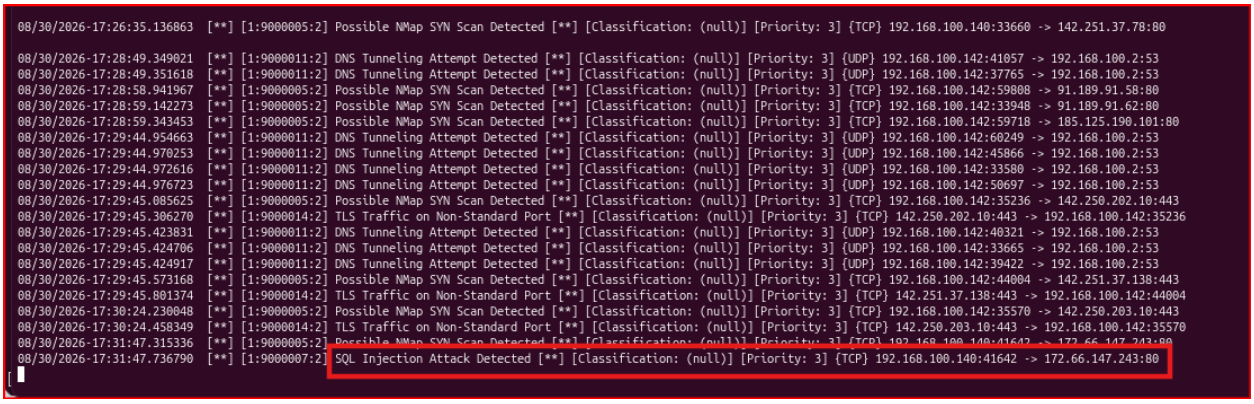
Attack 2: Suspicious User-Agent

- **Suricata Alert:** "Nmap User-Agent Detected"
- **Zeek Context:** HTTP request logged with User-Agent = Nmap Scripting Engine
- **Conclusion:** Malicious/suspicious tool signature detected in the User-Agent field.

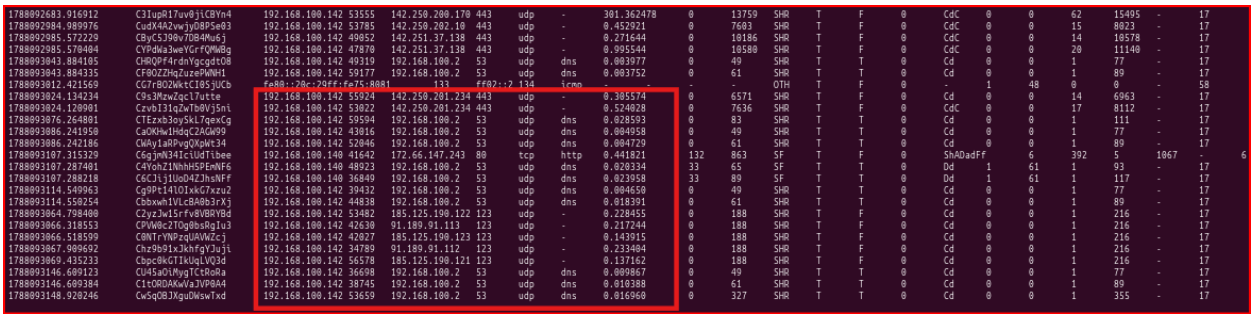
Attack 3: SQL Injection Style Request



- Suricata terminal (Terminal A)



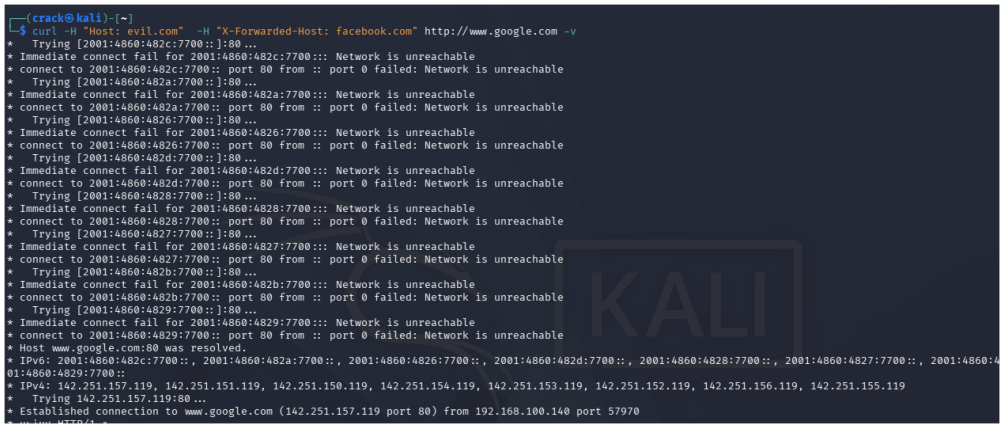
- Zeek terminal (Terminal B)



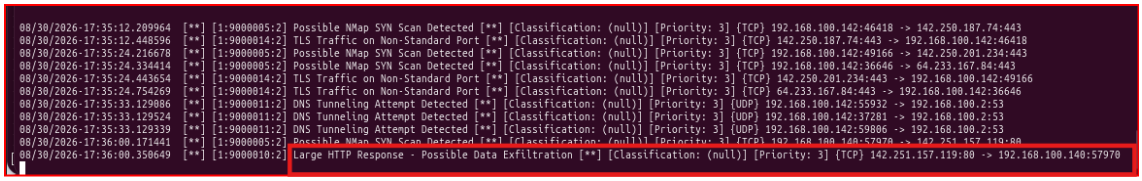
Attack 3: SQL Injection Attempt

- **Suricata Alert:** "SQL Injection Attempt Detected"
- **Zeek Context:** HTTP request containing SQL meta-characters and UNION/SELECT patterns
- **Conclusion:** True Positive. Attempt to inject SQL payload was successfully detected.

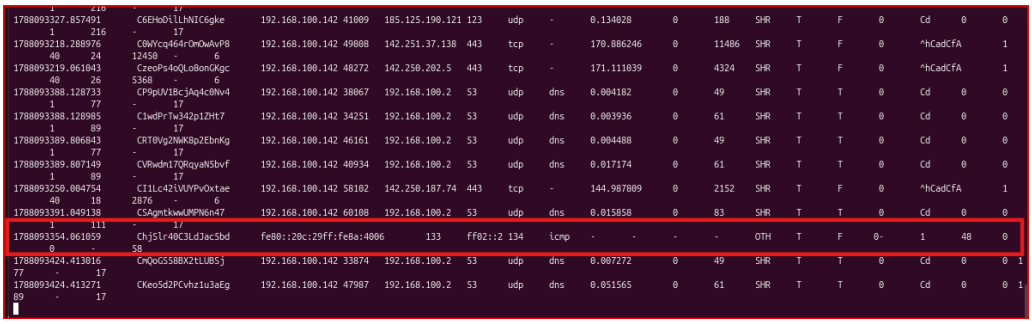
Attack 4: Suspicious Host Header



- Suricata terminal (Terminal A)



- Zeek terminal (Terminal B)



Attack 4: Suspicious Host Header

- **Suricata Alert:** Related policy / suspicious header alerts
- **Zeek Context:** HTTP request with modified Host header (facebook.com)
- **Conclusion:** Host header manipulation detected, commonly used in attacks and traffic redirection techniques.

Suricata provided real-time detection of reconnaissance, suspicious tooling, and injection attempts. Zeek supplied the full connection and application-layer context required for investigation. Together, both tools enabled complete visibility from attack execution to forensic analysis.

PART 4: MongoDB + RITA – Threat Hunting

MONGO DB Setup

We will use Docker for MongoDB 4.4 + RITA-legacy.

Step 1: Install Docker

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita-legacy$ sudo apt update
sudo apt install -y docker.io docker-compose
sudo systemctl start docker
sudo systemctl enable docker
sudo usermod -aG docker $USER
Hit:1 http://pk.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://pk.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 https://repo.mongodb.org/apt/ubuntu focal/mongodb-org/4.4 InRelease [4,009 B]
```

Step 2: Run MongoDB 4.4 in Docker

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita-legacy$ mkdir -p ~/mongodb-data
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita-legacy$ docker run -d \
  --name mongodb \
  --restart unless-stopped \
  -p 27017:27017 \
  -v ~/mongodb-data:/data/db \
  mongo:4.4
Unable to find image 'mongo:4.4' locally
4.4: Pulling from library/mongo
98279e2678fe: Pull complete
8fd00fb17365: Pull complete
130589ade9bb: Pull complete
13b7e930469f: Pull complete
6a696631c981: Pull complete
d103ead1acfe: Pull complete
1f21fe582607: Pull complete
c67f7c95021c: Pull complete
90e23540f057: Download complete
4e6a45bd2725: Download complete
Digest: sha256:4be76f674fc4b27859816811b8baa3c51830eb1dbf4ca81a51e26b79edd662ef
Status: Downloaded newer image for mongo:4.4
b0db1f313c1ac7b93e93a9ff7959c2a7899591ca4a70286888b7f0c18b52d074b
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita-legacy$
```

Step 3: Verify MongoDB is running

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita-legacy$
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita-legacy$ docker ps
docker logs mongodb --tail 20
CONTAINER ID   IMAGE      COMMAND                  CREATED          STATUS          PORTS                      NAMES
b0db1f313c1a   mongo:4.4  "docker-entrypoint.s…"   25 seconds ago   Up 24 seconds   0.0.0.0:27017->27017/tcp   mongodb
{"t":{"$date":"2026-08-30T20:33:27.599+00:00"},"s":"I",  "c":"STORAGE",  "id":4795906, "ctx":"initandlisten","msg":"WiredTiger opened","attr":{"durationMillis":584}}
{"t":{"$date":"2026-08-30T20:33:27.599+00:00"},"s":"I",  "c":"RECOVERY", "id":23987,   "ctx":"initandlisten","msg":"WiredTiger recoveryTimestamp","attr":{"recoveryTimestamp":{"$timestamp":{"t
:0,"i":0}}}}
{"t":{"$date":"2026-08-30T20:33:27.613+00:00"},"s":"I",  "c":"STORAGE",  "id":22262,   "ctx":"initandlisten","msg":"Timestamp monitor starting"}
{"t":{"$date":"2026-08-30T20:33:27.618+00:00"},"s":"W",  "c":"CONTROL",  "id":22120,   "ctx":"initandlisten","msg":"Access control is not enabled for the database. Read and write access to dat
a and configuration is unrestricted","tags":["startupWarnings"]}
{"t":{"$date":"2026-08-30T20:33:27.623+00:00"},"s":"W",  "c":"CONTROL",  "id":22184,   "ctx":"initandlisten","msg":"Soft rlimits too low","attr":{"currentValue":1024,"recommendedMinimum":64000
},"tags":["startupWarnings"]}
{"t":{"$date":"2026-08-30T20:33:27.630+00:00"},"s":"I",  "c":"STORAGE",  "id":20320,   "ctx":"initandlisten","msg":"createCollection","attr":{"namespace":"admin.system.version","uuidDispositio
n":"provided","uuid":{"uuid":{"$uuid":"b00edf76-077f-4a45-bd85-5e492945f2f6"}},"options":{"uuid":{"$uuid":"b00edf76-077f-4a45-bd85-5e492945f2f6"}}}}
{"t":{"$date":"2026-08-30T20:33:27.645+00:00"},"s":"I",  "c":"INDEX",    "id":20345,   "ctx":"initandlisten","msg":"Index build: done building","attr":{"buildUUID":null,"namespace":"admin.syst
em.version","index":"_id_","commitTimestamp":{"$timestamp":{"t":0,"i":0}}}}
{"t":{"$date":"2026-08-30T20:33:27.645+00:00"},"s":"I",  "c":"COMMAND",  "id":20459,   "ctx":"initandlisten","msg":"Setting featureCompatibilityVersion","attr":{"newVersion":"4.4"}}
{"t":{"$date":"2026-08-30T20:33:27.645+00:00"},"s":"I",  "c":"STORAGE",  "id":20536,   "ctx":"initandlisten","msg":"Flow Control is enabled on this deployment"}
{"t":{"$date":"2026-08-30T20:33:27.650+00:00"},"s":"I",  "c":"STORAGE",  "id":20320,   "ctx":"initandlisten","msg":"createCollection","attr":{"namespace":"local.startup_log","uuidDisposition":
"generated","uuid":{"uuid":{"$uuid":"ab5210a7-a3ec-478a-938c-d8c75b48375f"}},"options":{"capped":true,"size":10485760}}}
{"t":{"$date":"2026-08-30T20:33:27.673+00:00"},"s":"I",  "c":"INDEX",    "id":20345,   "ctx":"initandlisten","msg":"Index build: done building","attr":{"buildUUID":null,"namespace":"local.star
tup_log","index":"_id_","commitTimestamp":{"$timestamp":{"t":0,"i":0}}}}
{"t":{"$date":"2026-08-30T20:33:27.673+00:00"},"s":"I",  "c":"FTDC",     "id":20625,   "ctx":"initandlisten","msg":"Initializing full-time diagnostic data capture","attr":{"dataDirectory":"/da
ta/db/diagnostic.data"}}
{"t":{"$date":"2026-08-30T20:33:27.674+00:00"},"s":"I",  "c":"REPL",     "id":6015317, "ctx":"initandlisten","msg":"Setting new configuration state","attr":{"newState":"ConfigReplicationDisabl
ed","oldState":"ConfigPreStart"}}
{"t":{"$date":"2026-08-30T20:33:27.676+00:00"},"s":"I",  "c":"STORAGE",  "id":20320,   "ctx":"LogicalSessionCacheRefresh","msg":"createCollection","attr":{"namespace":"config.system.sessions",
"uuidDisposition":"generated","uuid":{"uuid":{"$uuid":"d6ef87d5-574f-4c9b-a5f6-1d35e52a52a2"}},"options":{}}}
{"t":{"$date":"2026-08-30T20:33:27.675+00:00"},"s":"I",  "c":"CONTROL",  "id":20712,   "ctx":"LogicalSessionCacheReap","msg":"Sessions collection is not set up; waiting until next sessions rea
p interval","attr":{"error":"NamespaceNotFound: config.system.sessions does not exist"}}
{"t":{"$date":"2026-08-30T20:33:27.676+00:00"},"s":"I",  "c":"NETWORK",  "id":23015,   "ctx":"listener","msg":"Listening on","attr":{"address":"/tmp/mongodb-27017.sock"}}
{"t":{"$date":"2026-08-30T20:33:27.676+00:00"},"s":"I",  "c":"NETWORK",  "id":23015,   "ctx":"listener","msg":"Listening on","attr":{"address":"0.0.0.0"}}
{"t":{"$date":"2026-08-30T20:33:27.677+00:00"},"s":"I",  "c":"NETWORK",  "id":23016,   "ctx":"listener","msg":"Waiting for connections","attr":{"port":27017,"ssl":"off"}}
{"t":{"$date":"2026-08-30T20:33:27.698+00:00"},"s":"I",  "c":"INDEX",    "id":20345,   "ctx":"LogicalSessionCacheRefresh","msg":"Index build: done building","attr":{"buildUUID":null,"namespace
":"config.system.sessions","index":"_id_","commitTimestamp":{"$timestamp":{"t":0,"i":0}}}}
{"t":{"$date":"2026-08-30T20:33:27.698+00:00"},"s":"I",  "c":"INDEX",    "id":20345,   "ctx":"LogicalSessionCacheRefresh","msg":"Index build: done building","attr":{"buildUUID":null,"namespace
":"config.system.sessions","index":"lsidTTLIndex","commitTimestamp":{"$timestamp":{"t":0,"i":0}}}}
```

Install RITA (MongoDB version)

We will use an older stable version of RITA that still supports MongoDB.

1. Install Go (required to build RITA)

```
sudo apt install -y golang-go git make
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$ sudo apt install -y golang-go git make
[sudo: authenticate] Password:
golang-go is already the newest version (2:1.26~1).
git is already the newest version (1:2.53.0-1ubuntu1).
make is already the newest version (4.4.1-3).
Summary:
```

2. Clone the correct MongoDB-compatible RITA (rita-legacy)

```
git clone https://github.com/activecm/rita-legacy.git
```

```
cd rita-legacy
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita$ cd ~
rm -rf rita
operation-dept@operation-dept-VMware-Virtual-Platform:~$ git clone https://github.com/activecm/rita-legacy.git
Cloning into 'rita-legacy'...
remote: Enumerating objects: 7940, done.
remote: Counting objects: 100% (199/199), done.
remote: Compressing objects: 100% (128/128), done.
remote: Total 7940 (delta 129), reused 71 (delta 71), pack-reused 7741 (from 3)
Receiving objects: 100% (7940/7940), 15.57 MiB | 3.50 MiB/s, done.
Resolving deltas: 100% (5178/5178), done.
```

3. Checkout the latest stable MongoDB version

```
git checkout v4.8.2
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ cd rita-legacy
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita-legacy$ git checkout v4.8.2
Note: switching to 'v4.8.2'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:
```

4. Build RITA

```
make
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita-legacy$ make
go build -ldflags='-X github.com/activecm/rita-legacy/config.Version=v4.8.2 -X github.com/activecm/rita-legacy/config.ExactVersion=v4.8.2-0-g63829cb'
go: downloading github.com/urfave/cli v1.20.0
go: downloading github.com/activecm/mgosec v0.1.2-0.20191108195135-d918492993da
go: downloading github.com/creasty/defaults v1.3.0
go: downloading gopkg.in/yaml.v2 v2.2.2
go: downloading github.com/globalsign/mgo v0.0.0-20190517090918-73267e130ca1
go: downloading github.com/olekukonko/tablewriter v0.0.2-0.20190214164707-93462a5dfaa6
go: downloading github.com/sirupsen/logrus v1.4.2
go: downloading github.com/pbnjay/memory v0.0.0-20201129165224-b12e5d931931
go: downloading github.com/vbauerster/mpb v3.3.4+incompatible
go: downloading github.com/activecm/rita-bl v0.0.0-20220823191806-f014db21453d
go: downloading github.com/skratchdot/open-golang v0.0.0-20190104022628-a2dfa6d0dab6
go: downloading github.com/activecm/mgorus v0.1.1
go: downloading github.com/rifflock/lfshook v0.0.0-20180920164130-b9218ef580f5
go: downloading github.com/google/go-querystring v1.0.0
go: downloading github.com/mattn/go-runewidth v0.0.4
go: downloading golang.org/x/sys v0.0.0-20190422165155-953cdadca894
go: downloading github.com/json-iterator/go v1.1.11
```

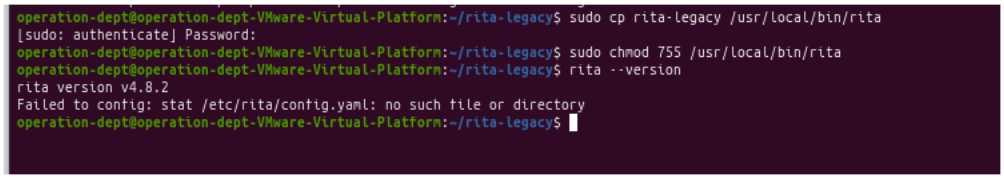
5. Install the binary

```
sudo cp rita /usr/local/bin/
```

```
sudo chmod 755 /usr/local/bin/rita
```

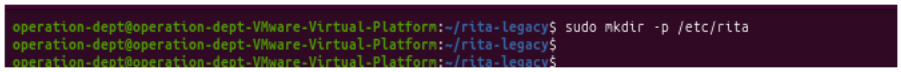
Verify

```
 rita -version
```

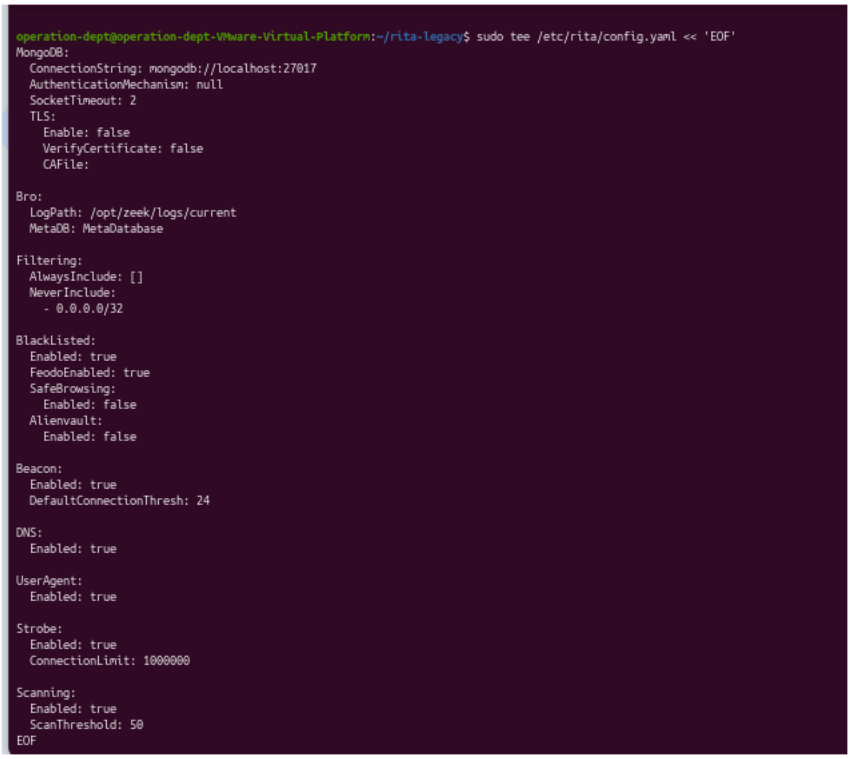


Create RITA Configuration

1. Create config directory



2. Create the configuration file for MongoDB



Fix ownership and permissions

```
sudo chown $USER:$USER /etc/rita/config.yaml

sudo chmod 644 /etc/rita/config.yaml
```



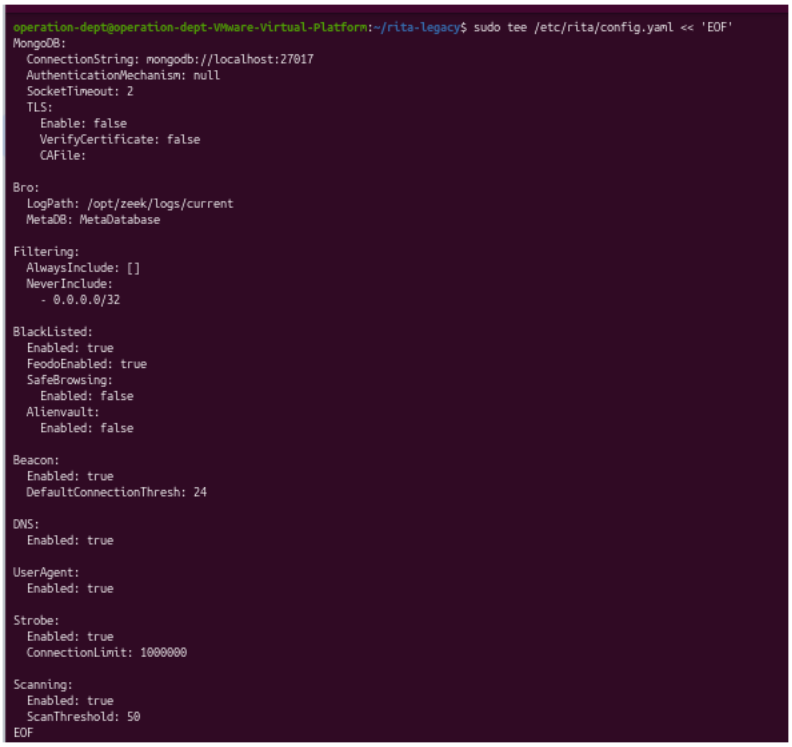
Also fix the directory

```
sudo chown $USER:$USER /etc/rita

sudo chmod 755 /etc/rita
```



Update RITA configuration



Test RITA connection

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita-legacy$ docker ps
CONTAINER ID   IMAGE       COMMAND                  CREATED         STATUS         PORTS                      NAMES
b0db1f313c1a   mongo:4.4   "docker-entrypoint.s…"   2 minutes ago   Up 2 minutes   0.0.0.0:27017->27017/tcp   mongodb
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita-legacy$
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/rita-legacy$ rita test-config

UserConfig:
  UpdateCheckFrequency: 14
MongoDB:
  ConnectionString: mongodb://localhost:27017
  AuthenticationMechanism: ""
  SocketTimeout: 2h0m0s
  TLS:
    Enable: false
    VerifyCertificate: false
    CAFile: ""
  MetaDB: MetaDatabase
Rolling:
  DefaultChunks: 24
  rolling: false
  currentchunk: 0
  totalchunks: 0
LogConfig:
  LogLevel: 2
  RitaLogPath: /var/lib/rita/logs
  LogToFile: true
  LogToDB: true
BlackListed:
  Enabled: true
```

Make sure Zeek is running and producing logs

```
cd /opt/zeek

sudo bin/zeekctl status

ls -la /opt/zeek/logs/current/
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek$ sudo bin/zeekctl status
Name         Type       Host          Status    Pid    Started
zeek         standalone localhost     running   11590  31 Aug 01:42:58
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek$ ls -la /opt/zeek/logs/current/
total 112
drwxr-xr-x 2 root root  4096 Aug 31 01:43 .
drwxr-xr-x 7 root root  4096 Aug 31 01:44 ..
-rw------- 1 root root   129 Aug 31 01:42 .cmdline
-rw------- 1 root root   362 Aug 31 01:42 .env_vars
-rw------- 1 root root     6 Aug 31 01:42 .pid
-rw------- 1 root root    61 Aug 31 01:42 .startup
-rwx------ 1 root root    19 Aug 31 01:43 .status
-rw------- 1 root root  1832 Aug 31 01:44 conn.log
-rw------- 1 root root  1586 Aug 31 01:43 dns.log
-rw------- 1 root root   242 Aug 31 01:43 known_services.log
-rw------- 1 root root 35179 Aug 31 01:43 loaded_scripts.log
-rw------- 1 root root   251 Aug 31 01:43 packet_filter.log
-rw------- 1 root root   674 Aug 31 01:43 reporter.log
-rw------- 1 root root   708 Aug 31 01:43 stats.log
-rw------- 1 root root    20 Aug 31 01:43 stderr.log
-rw------- 1 root root   204 Aug 31 01:42 stdout.log
-rw------- 1 root root  9611 Aug 31 01:43 telemetry.log
-rw------- 1 root root   704 Aug 31 01:43 weird.log
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek$
```

Fix RITA directories and permissions (optional)

```
sudo mkdir -p /var/lib/rita/logs /var/lib/rita/databases

sudo chown -R $USER:$USER /var/lib/rita

sudo chmod -R 755 /var/lib/rita
```

```
orm:~$ sudo mkdir -p /var/lib/rita/logs /var/lib/rita/databases
orm:~$ sudo chown -R $USER:$USER /var/lib/rita
orm:~$ sudo chmod 755 /var/lib/rita
orm:~$
```

1. Give your user read access to the current logs

```
sudo chown -R $USER:$USER /opt/zeek/logs/current/

sudo chmod -R 644 /opt/zeek/logs/current/*
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek$ sudo chown -R $USER:$USER /opt/zeek/logs/current/
sudo chmod -R 644 /opt/zeek/logs/current/*
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek$
```

2. Verify you can now read the logs

```
ls -la /opt/zeek/logs/current/
```

```
head -n 3 /opt/zeek/logs/current/conn.log
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek$ ls -la /opt/zeek/logs/current/
head -n 3 /opt/zeek/logs/current/conn.log
total 384
drwxr-xr-x 2 operation-dept operation-dept   4096 Aug 31 01:48 .
drwxr-xr-x 7 root           root             4096 Aug 31 01:50 ..
-rw------- 1 operation-dept operation-dept    129 Aug 31 01:42 .cmdline
-rw------- 1 operation-dept operation-dept    362 Aug 31 01:42 .env_vars
-rw------- 1 operation-dept operation-dept      6 Aug 31 01:42 .pid
-rw------- 1 operation-dept operation-dept     61 Aug 31 01:42 .startup
-rwx------ 1 operation-dept operation-dept     19 Aug 31 01:43 .status
-rw-r--r-- 1 operation-dept operation-dept    250 Aug 31 01:44 capture_loss.log
-rw-r--r-- 1 operation-dept operation-dept  41069 Aug 31 01:52 conn.log
-rw-r--r-- 1 operation-dept operation-dept  34374 Aug 31 01:52 dns.log
-rw-r--r-- 1 operation-dept operation-dept    781 Aug 31 01:48 files.log
-rw-r--r-- 1 operation-dept operation-dept   1140 Aug 31 01:51 http.log
-rw-r--r-- 1 operation-dept operation-dept    242 Aug 31 01:43 known_services.log
-rw-r--r-- 1 operation-dept operation-dept  35179 Aug 31 01:43 loaded_scripts.log
-rw-r--r-- 1 operation-dept operation-dept    753 Aug 31 01:44 notice.log
-rw-r--r-- 1 operation-dept operation-dept    251 Aug 31 01:43 packet_filter.log
-rw-r--r-- 1 operation-dept operation-dept   5244 Aug 31 01:51 quic.log
-rw-r--r-- 1 operation-dept operation-dept    674 Aug 31 01:43 reporter.log
-rw-r--r-- 1 operation-dept operation-dept   3474 Aug 31 01:52 ssl.log
-rw-r--r-- 1 operation-dept operation-dept    817 Aug 31 01:48 stats.log
-rw-r--r-- 1 operation-dept operation-dept     20 Aug 31 01:43 stderr.log
-rw-r--r-- 1 operation-dept operation-dept    204 Aug 31 01:42 stdout.log
-rw-r--r-- 1 operation-dept operation-dept 169552 Aug 31 01:52 telemetry.log
-rw-r--r-- 1 operation-dept operation-dept   4843 Aug 31 01:52 weird.log
#separator \x09
```

3. Import into RITA

```
cd ~
```

```
rita import /opt/zeek/logs/current zeek_lab
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ rita import /opt/zeek/logs/current zeek_lab

        [+] Importing [/opt/zeek/logs/current]:
        [-] Verifying log files have not been previously parsed into the target dataset ...
        [-] Processing batch 1 of 1
        [-] Parsing logs to: zeek_lab ...
        [-] Parsing /opt/zeek/logs/current/conn.log -> zeek_lab
        [-] Parsing /opt/zeek/logs/current/dns.log -> zeek_lab
        [-] Parsing /opt/zeek/logs/current/http.log -> zeek_lab
        [-] Parsing /opt/zeek/logs/current/ssl.log -> zeek_lab
        [-] Finished parsing logs in 10ms
        [-] Host Analysis:            36 / 36  [==================] 100 %
        [-] Unique Connection Analysis: 35 / 35  [==================] 100 %
        [-] Unique Connection Aggregation: 1 / 1  [==================] 100 %
        [!] No Proxy Uconn data to analyze
        [!] No TLS or HTTP connections to analyze
        [-] Exploded DNS Analysis:    45 / 45  [==================] 100 %
        [-] Hostname Analysis:        45 / 45  [==================] 100 %
        [-] Beacon Analysis:          35 / 35  [==================] 100 %
        [-] Beacon Aggregation:       1 / 1  [==================] 100 %
        [!] No Proxy Beacon data to analyze
        [!] No TLS or HTTP Beacon data to analyze
        [-] UserAgent Analysis:       2 / 2  [==================] 100 %
        [-] UserAgent Aggregation:    1 / 1  [==================] 100 %
        [!] No invalid certificate data to analyze
        [-] Indexing log entries ...
```

4. Check results

```
rita list
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ rita list
zeek_lab
operation-dept@operation-dept-VMware-Virtual-Platform:~$
```

Check the summary and analysis results

```
rita show-long-connections zeek_lab
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ rita show-long-connections zeek_lab
Source IP,Destination IP,Port:Protocol:Service,Total Duration,Longest Duration,Connections,Total Bytes,State
192.168.100.142,142.251.37.138,443:tcp:- 443:udp:-,750.062,171.03,13,89844,closed
192.168.100.142,142.250.200.170,443:tcp:- 443:udp:-,734.234,170.969,14,143162,closed
192.168.100.142,142.250.203.170,443:udp:- 443:tcp:-,514.359,171.209,8,62006,closed
192.168.100.142,151.101.141.91,443:tcp:-,459.13,170.722,6,15529,closed
192.168.100.142,142.250.187.78,443:tcp:- 443:udp:-,344.609,170.919,6,183631,closed
192.168.100.142,142.250.187.69,443:tcp:-,342.261,171.648,4,18064,closed
192.168.100.142,173.194.76.84,443:tcp:- 443:udp:-,342.08,170.856,7,20690,closed
192.168.100.142,142.250.202.5,443:tcp:-,323.677,323.677,2,12818,closed
192.168.100.142,34.107.243.93,443:tcp:-,298.95,298.95,3,224,closed
192.168.100.142,142.250.201.234,443:udp:- 443:tcp:-,209.862,203.709,4,33600,closed
192.168.100.142,142.250.187.33,443:tcp:- 443:udp:-,199.831,171.016,3,24439,closed
192.168.100.142,142.250.202.78,443:tcp:- 443:udp:-,199.294,170.3,4,44110,closed
192.168.100.142,142.250.187.74,443:tcp:- 443:udp:-,193.154,170.496,7,57230,closed
192.168.100.142,34.54.185.247,443:tcp:- 443:udp:-,180.407,170.568,3,11710,closed
192.168.100.142,142.250.202.14,443:tcp:- 443:udp:-,174.361,170.51,3,96381,closed
192.168.100.142,142.250.202.106,443:udp:- 443:tcp:-,172.423,171.066,4,35006,closed
192.168.100.142,142.250.202.138,443:tcp:- 443:udp:-,172.184,170.803,4,34194,closed
192.168.100.142,142.250.200.174,443:tcp:-,171.013,171.013,2,7818,closed
192.168.100.142,142.250.202.10,443:tcp:-,170.945,170.945,2,12489,closed
192.168.100.142,142.251.37.67,443:tcp:-,170.18,170.18,2,2753,closed
192.168.100.142,142.250.202.3,443:udp:-,61.14,60.655,3,20513,closed
operation-dept@operation-dept-VMware-Virtual-Platform:~$
```

check how many connections exist

Quick check of log size

```
wc -l /opt/zeek/logs/current/conn.log
```

```
head -n 5 /opt/zeek/logs/current/conn.log
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ wc -l /opt/zeek/logs/current/conn.log
450 /opt/zeek/logs/current/conn.log
operation-dept@operation-dept-VMware-Virtual-Platform:~$ head -n 5 /opt/zeek/logs/current/conn.log
#separator \x09
#set_separator  ,
#empty_field    (empty)
#unset_field    -
#path   conn
operation-dept@operation-dept-VMware-Virtual-Platform:~$
```

PART 5: Final Testing, Validation & End-to-End Verification

Generate 4 Massive Attacks (from Kali)

Attack 1: Strong Beaconing

```
┌──(crack㉿kali)-[~]
└─$ echo "══ Starting Attack 1: Beaconing ══"
for i in {1..40}; do
  curl -s -A "Mozilla/5.0 (Windows NT 10.0; Win64; x64)" http://192.168.100.142/ > /dev/null
  sleep 6
done
echo "══ Attack 1 Completed ══"
══ Starting Attack 1: Beaconing ══
```

Attack 2: Aggressive Port Scan

```
┌──(crack㉿kali)-[~]
└─$ echo "══ Starting Attack 2: Port Scan ══"
nmap -sS -T4 -p- --min-rate 800 192.168.100.142
echo "══ Attack 2 Completed ══"
══ Starting Attack 2: Port Scan ══
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-31 10:29 +0500
Nmap scan report for 192.168.100.142 (192.168.100.142)
Host is up (0.00098s latency).
Not shown: 65533 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
27017/tcp open  mongod
MAC Address: 00:0C:29:8A:40:06 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 87.41 seconds
══ Attack 2 Completed ══
```

Attack 3: DNS Anomaly / Tunneling Style

```
┌──(crack㉿kali)-[~]
└─$ echo "══ Starting Attack 4: DNS Anomaly ══"
for i in {1..35}; do
  dig @8.8.8.8 $(head /dev/urandom | tr -dc a-z0-9 | head -c 50).evil-c2.com
done
echo "══ Attack 4 Completed ══"
══ Starting Attack 4: DNS Anomaly ══

; <<>> DiG 9.20.26-1-Debian <<>> @8.8.8.8 3m110vk3y2v9kdm4pfe5i8jfiyvekndcx9oz0p2pyc32kjqqo1.evil-c2.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; →→HEADER←← opcode: QUERY, status: NXDOMAIN, id: 35482
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;3m110vk3y2v9kdm4pfe5i8jfiyvekndcx9oz0p2pyc32kjqqo1.evil-c2.com.        IN A

;; AUTHORITY SECTION:
com.                    900     IN      SOA     a.gtld-servers.net. nstld.verisign-grs.com. 1788154190 18
00 900 604800 900

;; Query time: 147 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Mon Aug 31 10:30:26 PKT 2026
;; MSG SIZE  rcvd: 164
```

Test 1: Check Suricata Alerts

```
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$ sudo cat /var/log/suricata/eve.json | jq -r 'select(.event_type=="alert") | .alert.signature' | sort | uni
q -c | sort -rn | head -20
jq: parse error: Invalid numeric literal at line 6728, column 2953
   2107 Possible NMap SYN Scan Detected
     55 DNS Tunneling Attempt Detected
     54 TLS Traffic on Non-Standard Port
      2 NMap HTTP Probe Detected
      2 Known Malware User-Agent Detected
      1 SQL Injection Attack Detected
      1 Large HTTP Response - Possible Data Exfiltration
      1 Facebook Access Blocked
operation-dept@operation-dept-VMware-Virtual-Platform:~/zeek-build/zeek-8.0.10/build$
```

Test 2: Check Zeek Logs

```
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek/logs/current$ zeek-cut id.orig_h id.resp_h id.resp_p service duration < conn.log | tail -30
zeek-cut: skipping log line (not enough fields)
192.168.100.140 192.168.100.142 80      -       2.016129
fe80::20c:29ff:fe8a:4006        ff02::2 134     -       -
192.168.100.140 192.168.100.142 80      -       -
192.168.100.140 142.250.202.131 443     quic,ssl        0.320931
192.168.100.142 142.251.127.84  443     -       95.484709
192.168.100.140 142.250.202.106 443     quic,ssl        0.384132
192.168.100.140 192.168.100.142 80      -       -
192.168.100.142 142.250.203.170 443     -       -
192.168.100.140 192.168.100.2   53      dns     0.016958
192.168.100.140 192.168.100.2   53      dns     0.004149
192.168.100.142 192.168.100.2   53      dns     0.019852
192.168.100.142 192.168.100.2   53      dns     0.018736
192.168.100.140 142.250.202.131 443     quic,ssl        0.331947
192.168.100.140 192.168.100.142 80      -       -
192.168.100.142 192.168.100.2   53      dns     0.016125
192.168.100.142 192.168.100.2   53      dns     0.035215
192.168.100.142 192.168.100.2   53      dns     0.033382
192.168.100.140 142.250.202.78  443     quic,ssl        71.373394
192.168.100.140 192.168.100.2   53      dns     0.019438
192.168.100.140 192.168.100.2   53      dns     0.019421
192.168.100.142 192.168.100.2   53      dns     0.003309
192.168.100.142 192.168.100.2   53      dns     0.015767
192.168.100.142 192.168.100.2   53      dns     0.016540
192.168.100.140 192.168.100.2   53      dns     0.018024
192.168.100.140 192.168.100.2   53      dns     0.005716
192.168.100.142 192.168.100.2   53      dns     0.025078
192.168.100.142 192.168.100.2   53      dns     0.024861
fe80::20c:29ff:fe75:8081        ff02::2 134     -       -
192.168.100.142 192.168.100.2   53      dns     0.021116
192.168.100.142 192.168.100.2   53      dns     0.020865
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek/logs/current$
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek/logs/current$ zeek-cut query qtype_name < dns.log | tail -20
zeek-cut: skipping log line (not enough fields)
ssl.gstatic.com HTTPS
play.google.com -
play.google.com -
connectivity-check.ubuntu.com   -
chat.google.com A
chat.google.com HTTPS
play.google.com A
play.google.com HTTPS
ssl.gstatic.com -
ssl.gstatic.com -
ssl.gstatic.com A
ssl.gstatic.com HTTPS
prod-dynamite-prod-04-us-signaler-pa.clients6.google.com        -
prod-dynamite-prod-04-us-signaler-pa.clients6.google.com        -
-       -
signaler-pa.clients6.google.com -
-       -
signaler-pa.clients6.google.com -
5-prod-dynamite-prod-04-us-signaler-pa.clients6.google.com      -
5-prod-dynamite-prod-04-us-signaler-pa.clients6.google.com      -
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek/logs/current$
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:/opt/zeek/logs/current$ zeek-cut method host uri user_agent < http.log | tail -20
zeek-cut: skipping log line (not enough fields)
-       -       -       -
GET     clients2.google.com     /time/1/current?cup2key=10:flr9OAOOhf4JrhFVG0P7Nrt0n3RzWZqVvIQn0Ez04H4&cup2hreq=e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855    Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36
GET     edgedl.me.gvt1.com      /edgedl/diffgen-puffin/gonjkmgdlgnkkcocnoeinlnaijmmjnil/d9bf7c9e3e39c8833d939c0050d2833094614015afccc3e9ab4e80f0974684f00      Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36
GET     edgedl.me.gvt1.com      /edgedl/diffgen-puffin/lnelglejhenejginpboagddgdfbepgmp/dfad60ca5745961a1c227c495b275c80e95512abc7041821f083e903c59f9c49      Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36
GET     edgedl.me.gvt1.com      /edgedl/diffgen-puffin/hfnkpimlhhgieaddgfemjhofmfblmnib/f882aed27020a09763e77299b7e3aa3b65aa3212326fa993f4fdd7e2f8683c5c      Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36
GET     edgedl.me.gvt1.com      /edgedl/diffgen-puffin/efniojlnjndmcbiieegkicadnoecjjef/39cb25e08e81c8c00fddeed238e0ef4500782e22dd6548d0dd0469f4a4f74e8d      Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36
GET     edgedl.me.gvt1.com      /edgedl/diffgen-puffin/ggkkehgbnfjpeggfpleeakpidbkibbmn/a78665e5c964ec406137a2c7f4e7dc53073de1d31444dcdb070557f471cffef1      Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36
HEAD    www.google.com  /       curl/8.21.0
-       -       -       -
-       -       -       -
```

Test 4: Import & analyze in RITA

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ rita import /opt/zeek/logs/current final_lab

        [+] Importing [/opt/zeek/logs/current]:
        [-] Verifying log files have not been previously parsed into the target dataset ...
        [-] Processing batch 1 of 1
        [-] Parsing logs to: final_lab ...
        [-] Parsing /opt/zeek/logs/current/conn.log -> final_lab
        [-] Parsing /opt/zeek/logs/current/dns.log -> final_lab
        [-] Parsing /opt/zeek/logs/current/http.log -> final_lab
        [-] Parsing /opt/zeek/logs/current/ssl.log -> final_lab
        [-] Finished parsing logs in 211ms
        [-] Host Analysis:                 80 / 80  [==================] 100 %
        [-] Unique Connection Analysis: 102 / 102  [==================] 100 %
        [-] Unique Connection Aggregation: 2 / 2  [==================] 100 %
        [!] No Proxy Uconn data to analyze
        [-] SNI Connection Analysis:  37 / 37  [==================] 100 %
        [-] Exploded DNS Analysis:    167 / 167  [==================] 100 %
        [-] Hostname Analysis:        167 / 167  [==================] 100 %
        [-] Beacon Analysis:          102 / 102  [==================] 100 %
        [-] Beacon Aggregation:       2 / 2  [==================] 100 %
        [!] No Proxy Beacon data to analyze
        [-] SNI Beacon Analysis:      37 / 37  [==================] 100 %
        [-] SNI Beacon Aggregation:   2 / 2  [==================] 100 %
        [-] UserAgent Analysis:       4 / 4  [==================] 100 %
        [-] UserAgent Aggregation:    2 / 2  [==================] 100 %
        [!] No invalid certificate data to analyze
        [-] Indexing log entries ...
        [-] Updating metadatabase ...
        [-] Done!

Theres a new Major version of RITA 5.1.2 available at:
https://github.com/activecm/rita-legacy/releases

operation-dept@operation-dept-VMware-Virtual-Platform:~$
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ rita show-long-connections final_lab
Source IP,Destination IP,Port:Protocol:Service,Total Duration,Longest Duration,Connections,Total Bytes,State
192.168.100.142,142.250.187.46,443:tcp:- 443:udp:-,1329.38,1104.56,10,54464,closed
192.168.100.140,142.250.201.106,443:udp:quic,ssl 443:tcp:ssl,1209.38,833.422,8,147258,closed
192.168.100.142,142.250.202.138,443:udp:-,1048.02,1048.02,1,11529,closed
192.168.100.142,151.101.141.91,443:tcp:-,854.877,172.263,10,26375,closed
192.168.100.142,142.250.203.170,443:udp:- 443:tcp:-,740.202,172.556,14,55737,closed
192.168.100.142,142.250.202.131,443:udp:- 443:tcp:-,710.516,280.955,8,33788,closed
192.168.100.140,142.251.37.138,443:udp:quic,ssl 443:tcp:ssl,607.992,346.759,7,126139,closed
192.168.100.142,142.251.37.133,443:tcp:-,603.178,603.178,1,94421,closed
192.168.100.140,142.251.37.133,443:tcp:ssl,567.755,567.755,1,96600,closed
192.168.100.142,142.250.187.37,443:tcp:-,557.852,557.852,2,71549,closed
192.168.100.140,142.250.187.78,443:tcp:ssl 443:udp:quic,ssl,547.308,194.596,27,5763318,closed
192.168.100.140,142.250.203.10,443:udp:quic,ssl 443:tcp:ssl,543.216,300.75,4,71843,closed
192.168.100.142,142.251.127.84,443:udp:- 443:tcp:-,479.572,171.793,9,33163,closed
192.168.100.142,34.107.243.93,443:tcp:-,471.086,298.971,9,8416,closed
192.168.100.142,142.251.37.67,443:udp:- 443:tcp:-,412.904,241.848,5,20513,closed
192.168.100.142,142.250.187.42,443:udp:- 443:tcp:-,397.256,171.181,8,46705,closed
192.168.100.142,142.250.200.170,443:udp:- 443:tcp:-,397.09,170.921,8,59030,closed
192.168.100.142,142.250.187.74,443:udp:- 443:tcp:-,376.909,170.561,15,89838,closed
192.168.100.140,142.250.187.46,443:tcp:ssl 443:udp:quic,ssl 80:tcp:http,355.063,230.676,9,387405,closed
192.168.100.142,140.248.129.91,443:tcp:-,342.756,171.049,10,16163,closed
192.168.100.140,142.250.187.37,443:tcp:ssl,302.789,258.031,2,9638402,closed
192.168.100.140,142.250.200.170,443:tcp:ssl 443:udp:quic,ssl,291.571,251.438,6,142850,closed
192.168.100.142,104.18.29.234,443:tcp:-,282.208,176.505,3,8684,closed
192.168.100.140,142.250.202.14,443:tcp:ssl 443:udp:quic,ssl,233.034,227.301,2,1124784,closed
192.168.100.142,142.250.202.5,443:tcp:-,223.613,112.306,2,876,closed
192.168.100.142,142.250.202.74,443:udp:- 443:tcp:-,172.057,170.81,4,38295,closed
192.168.100.142,142.250.202.10,443:udp:- 443:tcp:-,171.778,170.857,4,20450,closed
192.168.100.142,34.149.226.178,443:tcp:-,171.491,171.359,3,6288,closed
192.168.100.142,64.233.167.84,443:tcp:-,171.119,171.119,2,6060,closed
192.168.100.142,142.250.202.78,443:tcp:- 443:udp:-,163.781,111.608,2,17387,closed
192.168.100.142,142.251.37.74,443:tcp:-,152.885,99.9877,3,3116,closed
192.168.100.142,142.250.200.174,443:tcp:-,112.117,112.117,1,438,closed
192.168.100.142,34.54.185.247,443:tcp:-,111.578,111.439,2,652,closed
192.168.100.140,142.250.202.78,443:udp:quic,ssl,111.427,71.3734,9,198436,closed
192.168.100.142,142.250.203.174,443:tcp:-,111.12,111.12,1,438,closed
operation-dept@operation-dept-VMware-Virtual-Platform:~$
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ rita show-beacons final_lab
Score,Source IP,Destination IP,Connections,Avg. Bytes,Total Bytes,TS Score,DS Score,Dur Score,Hist Score,Top Intvl
0.917,192.168.100.142,91.189.91.112,25,216,5400,1,0.667,1,1,65
0.776,192.168.100.140,142.250.202.131,40,26789,1071577,0.9,0.641,0.827,0.734,31
0.747,192.168.100.140,142.250.187.78,27,213456,5763318,0.514,0.793,0.75,0.929,38
0.442,192.168.100.140,8.8.8.8,115,521,59992,1,0.768,0,0,0
operation-dept@operation-dept-VMware-Virtual-Platform:~$
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ rita show-exploded-dns final_lab
Domain,Unique Subdomains,Times Looked Up
example.net,40,40
malicious-domain.com,40,40
evil-c2.com,35,35
google.com,28,380
clients6.google.com,14,142
googleapis.com,4,26
gstatic.com,3,129
mozilla.com,2,3
mozilla.org,2,6
services.mozilla.com,2,3
play.google.com,1,74
f0r6kuibdxzblclijnaszx4c5zwf0g8bjddgi1bmux4xa86on4.evil-c2.com,1,1
taskassist-pa.clients6.google.com,1,2
mdm3b65s18xwf206z2zqsyqosz5gj8b17edhzo9752xos.malicious-domain.com,1,1
ng0vi1ye58v2julxq8urcssj2chq00hmz0ueappógyu4e.malicious-domain.com,1,1
snapcraft.io,1,2
qatksftydouqlnzvqwxgbilgtwjyrp.example.net,1,1
wzvljzlmjdsvjzkoyrazzocbpeezoe.example.net,1,1
k0yzf296uso6dpplq7ea2wnfdx08win2zlmx39wds4un6wte9f.evil-c2.com,1,1
vrpjdtwwnpjgxiarjfbgfdrhtmbunw.example.net,1,1
studio.workspace.google.com,1,2
7641pu2ptf0801lten3c5uvqyfodwrfddcwqyehnuw4i5.malicious-domain.com,1,1
accounts.google.com.pk,1,2
workspace.google.com,1,2
waa-pa.clients6.google.com,1,24
www.google.com,1,10
g0rf8s4a2j7odi7jd5k79ilv77jq8yl81wzr5hpqb3ori.malicious-domain.com,1,1
optimizationguide-pa.googleapis.com,1,20
kcx7wj5f3baxa0btcit1us27oarcd8smygkg0iiswk2gyl05gd.evil-c2.com,1,1
ads.mozilla.org,1,4
contacts.google.com,1,2
ts5hzg7ocx0v0kk81y9jmh1erlvyra1i3wb5ypbnf8a1b8nw3i.evil-c2.com,1,1
ztawxefc71uskpesi3tokhjc8to20xmhbsfvnkfuxkpwtgcwum.evil-c2.com,1,1
a56kven2h4x6lsbmmhop2ov0xwihkh3asu3jv4mvzmgk0.malicious-domain.com,1,1
mhaopuaxghnpczisnbphwcsrornbkx.example.net,1,1
yrgdxsw5rd1w5bl76lws3yk5qwzi8ag5bp8xqv7ookbh3.malicious-domain.com,1,1
googleadservices.com,1,2
01al6pkfk3p8ewnkpchs37zu4qmiush0wmm7io4pqaqz9.malicious-domain.com,1,1
```

```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ rita show-useragents final_lab
User Agent,Times Used
No JA3 hash generated,275
Empty user agent string,7
Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36,6
curl/8.21.0,1
operation-dept@operation-dept-VMware-Virtual-Platform:~$
```

RITA Findings Summary

Analysis	Result
Exploded DNS	evil-c2.com (35 subdomains) and malicious-domain.com (40 subdomains) clearly detected
Beacons	Several beacons detected (highest score 0.917)
Long Connections	Mostly external Google connections
User Agents	Mostly normal + one curl

MITRE ATT&CK Mapping

Attack	Description	MITRE ATT&CK Tactic	Technique ID	Technique Name	Detection Tool
Attack 1	Repeated HTTP connections (Beaconing)	Command and control	T1071.001	Application Layer Protocol: Web Protocols	RITA (Beacon Analysis), Zeek
Attack 1	Repeated connections with regular intervals	Command and control	T1573	Encrypted Channel (or regular C2 beaconing behavior)	RITA
Attack 2	Aggressive Nmap SYN Scan	Discovery	T1046	Network Service Discovery	Suricata
Attack 2	Port scanning activity	Reconnaissance	T1595.001	Active Scanning: Scanning IP Blocks	Suricata
Attack 4	High volume of random long subdomain queries	Command and control	T1071.004	Application Layer Protocol: DNS	Suricata + RITA
Attack 4	DNS Tunneling style activity	Exfiltration / C2	T1048.003	Exfiltration Over Alternative Protocol: Exfiltration Over Unencrypted Non-C2 Protocol (DNS)	Suricata (DNS Tunneling alerts) + RITA (Exploded DNS)

The simulated attacks mapped to multiple MITRE ATT&CK techniques across Reconnaissance, Discovery, and Command and Control tactics. Suricata provided strong signature-based detection for scanning and DNS tunneling indicators, while RITA successfully identified behavioral patterns related to beaconing and exploded DNS activity.

1. Key Achievements

- Successfully deployed a production-style NSM stack (Suricata + Zeek + RITA)
- Demonstrated real-time detection and behavioral hunting
- Correlated alerts across multiple tools
- Produced clear evidence of DNS anomaly and scanning activity

2. Lessons Learned

- Proper log permissions and service stability are critical
- Beacon detection quality depends on consistent timing and volume
- Combining signature-based and behavioral tools provides superior coverage

3. Future Improvements

- Add Arkime for full packet capture and session reconstruction
- Implement automated alerting (e.g., Slack/Email on high-severity Suricata alerts)
- Introducing custom Zeek scripts for environment-specific detection
- Deploy the stack using Docker Compose for better reproducibility

Conclusion

The implementation of Suricata, Zeek, and RITA in an integrated pipeline proved highly effective for both real-time threat detection and post-event threat hunting. Suricata excelled at identifying known malicious patterns (port scans and DNS tunneling indicators), while Zeek provided the necessary depth of network metadata. RITA successfully converted Zeek logs into actionable hunting insights, particularly in detecting DNS anomalies.

This lab confirms that a multi-tool NSM architecture significantly improves visibility and detection capability compared to using any single tool in isolation. The hands-on experience also highlighted practical considerations such as log permissions, service stability, and the importance of clean traffic generation for reliable analysis.

Overall, the objectives of the laboratory were fully achieved. The deployed stack is functional, the attack simulations produced measurable results, and the correlation between tools was successfully demonstrated.
